

```

// File: Sources/xlsOne/XlsOneApp.swift
import AppKit
import SwiftUI
import xlsOneCore
import xlsOneUI

@main
struct XlsOneApp: App {
    @NSApplicationDelegateAdaptor(WorkspaceAppDelegate.self) private var appDelegate

    init() {
        NSWindow.allowsAutomaticWindowTabbing = false
        LocaleManager.shared.applyToFoundation()
    }

    var body: some Scene {
        XlsOneWorkspaceScene()
    }
}

// File: Sources/xlsOneUI/WorkspaceScene.swift
import AppKit
import SwiftUI
import UniformTypeIdentifiers
import xlsOneCore
import xlsOneLicense

public struct XlsOneWorkspaceScene: Scene {
    @StateObject private var viewModel = AppViewModel()
    @StateObject private var licenseManager = LicenseManager.shared
    @ObservedObject private var localeManager = LocaleManager.shared

    public init() {}

    public var body: some Scene {
        WindowGroup {
            ContentView()
                .environmentObject(viewModel)
                .environmentObject(licenseManager)
                .environment(\.locale, localeManager.swiftUILocale)
                .frame(minWidth: 800, minHeight: 600)
                .onAppear {
                    // Menu localization is handled in applicationDidFinishLaunching
                }
                .background(WorkspaceWindowTabDisabler())
        }
        .commands {
            WorkspaceCommands(viewModel: viewModel, licenseManager: licenseManager, localeManager: localeManager)
        }
    }
}

@objc private final class AppAboutHelpHandler: NSObject {
    static let shared = AppAboutHelpHandler()

    @objc func showCustomAbout() {
        DispatchQueue.main.async {
            WorkspaceAppPresentation.showAboutPanel()
        }
    }
}

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}
}

private struct WorkspaceCommands: Commands {
    @ObservedObject var viewModel: AppViewModel
    @ObservedObject var licenseManager: LicenseManager
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        WorkspaceFileCommands(viewModel: viewModel, localeManager: localeManager)
        WorkspaceEditCommands(viewModel: viewModel, localeManager: localeManager)
        WorkspaceAdjustmentMemoryCommands(viewModel: viewModel, localeManager: localeManager)
        WorkspaceViewCommands(localeManager: localeManager)
        WorkspaceWindowCommands(localeManager: localeManager)
        WorkspaceLicenseCommands(licenseManager: licenseManager, localeManager: localeManager)
        WorkspaceLanguageCommands(localeManager: localeManager)
        WorkspaceHelpCommands(localeManager: localeManager)
    }
}

private struct WorkspaceFileCommands: Commands {
    @ObservedObject var viewModel: AppViewModel
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandGroup(replacing: .appInfo) {
            Button(LocaleManager.loc("关于 表表归一")) {
                WorkspaceAppPresentation.showAboutPanel()
            }
        }

        CommandGroup(replacing: .newItem) {
            Button(LocaleManager.loc("新建批次")) {
                viewModel.closeAllFiles()
            }
            .keyboardShortcut("n", modifiers: .command)
        }

        CommandGroup(after: .newItem) {
            Divider()

            Button(LocaleManager.loc("导入文件...")) {
                viewModel.showOpenFileDialog()
            }
            .keyboardShortcut("o", modifiers: .command)

            Button(LocaleManager.loc("追加文件...")) {
                viewModel.showAddFileDialog()
            }
            .keyboardShortcut("o", modifiers: [.command, .shift])
            .disabled(!viewModel.toolbarPresentation.appendEnabled)
        }

        CommandGroup(replacing: .saveItem) {
            Button(LocaleManager.loc("导出 XLSX...")) {
                viewModel.exportResult()
            }
            .keyboardShortcut("s", modifiers: .command)
            .disabled(!viewModel.canExport)
        }
    }
}

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CommandGroup(after: .saveItem) {
    Divider()

    Button(LocaleManager.loc("刷新")) {
        viewModel.reloadFiles()
    }
    .keyboardShortcut("r", modifiers: .command)
    .disabled(viewModel.selectedFilePaths.isEmpty)
}

}

private struct WorkspaceEditCommands: Commands {
    @ObservedObject var viewModel: AppViewModel
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandGroup(replacing: .undoRedo) {
            Button(LocaleManager.loc("撤销上一步")) {
                viewModel.undoLastOverride()
            }
            .keyboardShortcut("z", modifiers: .command)
            .disabled(!viewModel.canUndoOverride)

            Divider()

            Button(LocaleManager.loc("清除本批次调整")) {
                viewModel.clearOverrides()
            }
            .disabled(viewModel.correctionCount == 0)
        }

        CommandGroup(replacing: .pasteboard) {}
    }
}

private struct WorkspaceAdjustmentMemoryCommands: Commands {
    @ObservedObject var viewModel: AppViewModel
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandMenu(LocaleManager.loc("调整记忆")) {
            Button(LocaleManager.loc("查看当前调整记忆")) {
                viewModel.showSchemaManagerWindow()
            }
            .keyboardShortcut(",", modifiers: .command)
            .disabled(!viewModel.canManageAdjustmentMemory)

            Button(LocaleManager.loc("保存当前调整记忆")) {
                viewModel.saveCurrentAdjustmentMemory()
            }
            .disabled(!viewModel.canManageAdjustmentMemory)
        }
    }
}

private struct WorkspaceViewCommands: Commands {
    @ObservedObject var localeManager: LocaleManager

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var body: some Commands {
    CommandGroup(replacing: .toolbar) {}
}

private struct WorkspaceWindowCommands: Commands {
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandGroup(replacing: .windowArrangement) {
            Button(LocaleManager.loc("最小化")) {
                NSApp.keyWindow?.miniaturize(nil)
            }
            .keyboardShortcut("m", modifiers: .command)

            Button(LocaleManager.loc("缩放")) {
                NSApp.keyWindow?.zoom(nil)
            }

            Divider()

            Button(LocaleManager.loc("前置全部窗口")) {
                NSApp.arrangeInFront(nil)
            }
        }
    }
}

private struct WorkspaceLicenseCommands: Commands {
    @ObservedObject var licenseManager: LicenseManager
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandMenu(LocaleManager.loc("许可")) {
            Button(LocaleManager.loc("激活/导入许可证...")) {
                licenseManager.showActivationSheet = true
            }
            .disabled(licenseManager.licenseState == .activated)
        }
    }
}

private struct WorkspaceLanguageCommands: Commands {
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandMenu(LocaleManager.loc("Language_menu")) {
            ForEach(LocaleManager.AppLanguage.allCases) { language in
                Button {
                    // Save-only: UI and menus stay unchanged until next launch.
                    UserDefaults.standard.set(language.rawValue, forKey: "AppLanguage")
                    localeManager.applyToFoundation(for: language)
                    showRestartToast(for: language)
                } label: {
                    HStack {
                        Text(language.displayName)
                        if localeManager.currentLanguage == language {
                            Image(systemName: "checkmark")
                        }
                    }
                }
            }
        }
    }
}

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    }
    }
}

private struct WorkspaceHelpCommands: Commands {
    @ObservedObject var localeManager: LocaleManager

    var body: some Commands {
        CommandGroup(replacing: .help) {
            Button(LocaleManager.loc("检查更新")) {
                WorkspaceAppPresentation.showAppStoreUpdatePanel()
            }

            Divider()

            Button(LocaleManager.loc("使用帮助")) {
                WorkspaceAppPresentation.showHelpPanel()
            }

            Divider()

            Button(LocaleManager.loc("关于 表表归一")) {
                WorkspaceAppPresentation.showAboutPanel()
            }
        }
    }
}

private struct WorkspaceWindowTabDisabler: NSViewRepresentable {
    func makeNSView(context: Context) -> WorkspaceWindowTabDisablingView {
        WorkspaceWindowTabDisablingView()
    }

    func updateNSView(_ nsView: WorkspaceWindowTabDisablingView, context: Context) {
        nsView.disableWindowTabs()
    }
}

private final class WorkspaceWindowTabDisablingView: NSView {
    override func viewDidMoveToWindow() {
        super.viewDidMoveToWindow()
        disableWindowTabs()
    }

    func disableWindowTabs() {
        DispatchQueue.main.async { [weak self] in
            guard let window = self?.window else { return }
            WorkspaceWindowTabController.disableTabs(for: window)
        }
    }
}

private enum WorkspaceWindowTabController {
    @MainActor
    static func disableAutomaticTabbing() {
        NSWindow.allowsAutomaticWindowTabbing = false
    }
}

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@MainActor
static func disableTabs(for window: NSWindow) {
    disableAutomaticTabbing()
    window.title = WorkspaceAppPresentation.localizedCaption
    window.tabbingMode = .disallowed
    window.tabbingIdentifier = ""
    if window.tabbedWindows != nil {
        window.toggleTabBar(nil)
    }
}

enum WorkspaceDialogPresenter {
    @MainActor
    private static var ownerWindow: NSWindow? {
        NSApp.keyWindow ?? NSApp.mainWindow ?? NSApp.windows.first { $0.isVisible }
    }

    @MainActor
    private static func center(_ dialogWindow: NSWindow, over owner: NSWindow?) {
        guard let owner else {
            dialogWindow.center()
            return
        }

        let ownerFrame = owner.frame
        let dialogFrame = dialogWindow.frame
        let origin = NSPoint(
            x: ownerFrame.midX - dialogFrame.width / 2,
            y: ownerFrame.midY - dialogFrame.height / 2
        )
        dialogWindow setFrameOrigin(origin)
    }

    @MainActor
    @discardableResult
    static func runModal(_ panel: NSSavePanel) -> NSApplication.ModalResponse {
        let owner = ownerWindow
        panel.contentView?.layoutSubtreeIfNeeded()
        center(panel, over: owner)
        return panel.runModal()
    }

    @MainActor
    @discardableResult
    static func runAlert(_ alert: NSAlert) -> NSApplication.ModalResponse {
        let owner = ownerWindow
        alert.layout()
        center(alert.window, over: owner)
        DispatchQueue.main.async {
            center(alert.window, over: owner)
        }
        return alert.runModal()
    }

    @MainActor
    static func runAlert(title: String, message: String, style: NSAlert.Style = .warning) {
        let alert = NSAlert()
        alert.alertStyle = style
        alert.messageText = title
    }

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alert.informativeText = message
    alert.addButton(withTitle: LocaleManager.Loc("确定"))
    runAlert(alert)
}
}

@MainActor
func showRestartToast(for language: LocaleManager.AppLanguage) {
    NotificationCenter.default.post(name: .showRestartToast, object: language)
}

extension Notification.Name {
    static let showRestartToast = Notification.Name("showRestartToast")
}

public final class WorkspaceAppDelegate: NSObject, NSApplicationDelegate {
    public func applicationDidFinishLaunching(_ notification: Notification) {
        WorkspaceWindowTabController.disableAutomaticTabbing()
    }
}

enum WorkspaceAppPresentation {
    static var localizedCaption: String {
        isPreferredLanguageChinese ? "xlsOne 表表归一" : "xlsOne"
    }

    static var displayName: String {
        localizedCaption
    }

    static var bundleDisplayName: String {
        Bundle.main.object(forKey: "CFBundleDisplayName") as? String ??
        Bundle.main.object(forKey: "CFBundleName") as? String ??
        "xlsOne"
    }

    static var versionLabel: String {
        let version = Bundle.main.object(forKey: "CFBundleShortVersionString") as? String
        let build = Bundle.main.object(forKey: "CFBundleVersion") as? String

        switch (version, build) {
        case let (.some(version), .some(build)) where !version.isEmpty && !build.isEmpty && version != build:
            return "版本 \(version) \(build)"
        case let (.some(version), _):
            return "版本 \(version)"
        case let (_, .some(build)):
            return "构建 \(build)"
        default:
            return "版本信息不可用"
        }
    }

    private static var marketingVersion: String {
        Bundle.main.object(forKey: "CFBundleShortVersionString") as? String ?? "1.0"
    }

    private static var isPreferredLanguageChinese: Bool {
        LocaleManager.shared.currentLanguage.isChineseLike()
    }
}

```

```
@MainActor
```

```
static func showAboutPanel() {
    let alert = NSAlert()
    let isChinese = isPreferredLanguageChinese
    alert.messageText = isChinese ? "关于 表表归一" : "About xlsOne"
    alert.informativeText = isChinese ? ""
        表表归一 V\(\marketingVersion)
```

多张同格式 Excel 报表一键汇总

把多张格式一致的 Excel 表合成一份汇总表。金额、数量等能相加的数会自动合计；名称、编号等不该相加的信息，会保留各文件里最常见的共同前缀。

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"" : ""

xlsOne V\(\marketingVersion)

One-click summary for multiple identically-formatted Excel reports.

Merge multiple Excel sheets with identical layouts into a single summary sheet. Amounts, quantities, and other summable values are automatically totaled. Names, codes, and other non-summable information retain the most common values across all files.

Author: Wang Zhen

Tech: Swift / SwiftUI

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""

```
alert.addButton(withTitle: LocaleManager.loc("确定"))
WorkspaceDialogPresenter.runAlert(alert)
NSApp.activate(ignoringOtherApps: true)
```

```
}
```

```
@MainActor
```

```
static func showHelpPanel() {
    let alert = NSAlert()
    let isChinese = isPreferredLanguageChinese
    alert.messageText = isChinese ? "使用帮助" : "Help"
    alert.informativeText = isChinese ? ""
        表表归一 · 多张同格式 Excel 报表一键汇总
```

1. 导入文件

拖拽 .xlsx 或 .xls 文件到窗口，或点击 [文件] → [导入文件]

2. 切换工作表

点击顶部的 Sheet 标签可切换要查看的报表页

3. 查看汇总

- 金额、数量等能相加的数自动合计
- 名称、编号等信息保留最常见的共同前缀
- 结构不一致的工作表会跳过，并提示原因

4. 穿透查阅

点击单元格可查看各文件原始值

5. 导出结果

点击 [导出 XLSX] 保存汇总结果

6. 单元格修正

在右侧面板可将单元格手动指定为标签或求和

快捷键:

Command+O 导入文件
 Command+Shift+O 追加文件
 Command+S 导出
 Command+R 刷新
 Command+N 清空
 Command+Z 撤销修正

"" : ""

xls0ne • One-click Excel Report Merger

1. Import Files

Drag .xlsx or .xls files into the window, or click [File] → [Import Files]

2. Switch Sheets

Click the Sheet tabs at the top to switch between report pages

3. View Summary

- Amounts, quantities, and summable numbers are automatically totaled
- Names, codes retain the most common values
- Sheets with inconsistent structure are skipped with a reason

4. Source Drilldown

Click any cell to view its raw values from each source file

5. Export Results

Click [Export XLSX] to save the summary

6. Cell Correction

Manually override cell type to Label or Sum in the right panel

Shortcuts:

Cmd+O Import Files
 Cmd+Shift+O Add Files
 Cmd+S Export
 Cmd+R Refresh
 Cmd+N Clear
 Cmd+Z Undo Correction

""

alert.addButton(withTitle: LocaleManager.loc("确定"))

WorkspaceDialogPresenter.runAlert(alert)

NSApp.activate(ignoringOtherApps: true)

}

@MainActor

static func showAppStoreUpdatePanel() {

WorkspaceDialogPresenter.runAlert(

title: LocaleManager.loc("检查更新"),

message: LocaleManager.loc("App Store 版本请通过 Mac App Store 获取更新。"),

style: .informational

)

NSApp.activate(ignoringOtherApps: true)

}

}

/// 应用状态管理

@MainActor

class AppViewModel: ObservableObject {

```

@Published var files: [ExcelFile] = []
@Published var selectedFilePaths: [String] = []
@Published var availableSheets: [String] = []
@Published var mergedResult: MergedResult?
@Published var validationReport: WorkbookValidationReport?
@Published var workspacePhase: WorkspacePhase = .idle
@Published var selectedCell: CellPosition?
@Published var anomalyQueue: [CellAnomalyItem] = []
@Published var allAnomalies: [CellAnomalyItem] = []
@Published var isLoading = false
@Published var errorMessage: String?
@Published var showError = false
@Published var selectedSheetSelection: WorkspaceSheetSelection?

private let parser = ExcelParser()
private let validator = WorkbookValidator()
private let smartMerger = SmartMerger()
private var mergedResultsBySheet: [String: MergedResult] = [:]

/// 显示打开文件对话框
func showOpenFileDialog() {
    let panel = NSOpenPanel()
    panel.allowsMultipleSelection = true
    panel.canChooseDirectories = false
    panel.canChooseFiles = true
    panel.allowedContentTypes = [
        UTType(filenameExtension: "xlsx")!,
        UTType(filenameExtension: "xls")!
    ]

    if WorkspaceDialogPresenter.runModal(panel) == .OK {
        let paths = panel.urls.map { $0.path }
        loadFiles(at: paths, append: !selectedFilePaths.isEmpty)
    }
}

/// 显示添加文件对话框
func showAddFileDialog() {
    let panel = NSOpenPanel()
    panel.allowsMultipleSelection = true
    panel.canChooseDirectories = false
    panel.canChooseFiles = true
    panel.allowedContentTypes = [
        UTType(filenameExtension: "xlsx")!,
        UTType(filenameExtension: "xls")!
    ]

    if WorkspaceDialogPresenter.runModal(panel) == .OK {
        let paths = panel.urls.map { $0.path }
        loadFiles(at: paths, append: true)
    }
}

/// 加载文件
func loadFiles(at paths: [String], append: Bool) {
    selectedFilePaths = append
        ? Self.deduplicatedPaths(selectedFilePaths + paths)
        : Self.deduplicatedPaths(paths)
    revalidateSelection()
}

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/// 关闭所有文件
func closeAllFiles() {
    files.removeAll()
    selectedFilePaths.removeAll()
    availableSheets.removeAll()
    mergedResult = nil
    validationReport = nil
    workspacePhase = .idle
    mergedResultsBySheet.removeAll()
    selectedCell = nil
    anomalyQueue.removeAll()
    allAnomalies.removeAll()
    matchedSchema = nil
    workspaceRuleState = .none
    userOverrides.removeAll()
    forgottenOverrideKeys.removeAll()
    overrideHistory.removeAll()
    workspaceBaseOverrides.removeAll()
    workspaceBaseSchemaID = nil
    workspaceActiveSchemaID = nil
    selectedSheetSelection = nil
}

/// 重新加载文件
func reloadFiles() {
    revalidateSelection()
}

/// 重置
func reset() {
    closeAllFiles()
}

/// 切换工作表
func switchToSheet(_ sheetName: String) {
    selectedSheetSelection = .mergeable(sheetName)
    syncSelectedSheetPresentation()
    refreshSelectionAndAnomalies()
}

/// 切换到跳过的工作表
func switchToSkippedSheet(_ sheetName: String) {
    selectedSheetSelection = .skipped(sheetName)
    syncSelectedSheetPresentation()
    refreshSelectionAndAnomalies()
}

/// 当前用户自定义的单元格覆盖
@Published var userOverrides: [CellTypeOverride] = []

/// 覆盖操作历史（用于撤销）
private var overrideHistory: [OverrideOperation] = []

/// 本批次明确要求移除的已记住调整
private var forgottenOverrideKeys: Set<StoredOverrideKey> = []

/// 当前工作区在本批次开始前的已记住调整快照
private var workspaceBaseOverrides: [CellTypeOverride] = []
private var workspaceBaseSchemaID: UUID?

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private var workspaceActiveSchemaID: UUID?

/// 是否显示规则管理器
@Published var showSchemaManager = false

/// 当前匹配的规则
@Published var matchedSchema: MergeSchema?

/// 当前工作区的规则应用状态
@Published var workspaceRuleState: WorkspaceRuleState = .none

var canManageAdjustmentMemory: Bool {
    workspacePhase == .ready && !files.isEmpty && !availableSheets.isEmpty
}

var currentAdjustmentMemory: MergeSchema? {
    matchedSchema
}

var canExport: Bool {
    workspacePhase == .ready &&
    validationReport?.readiness == .ready &&
    !files.isEmpty &&
    !mergedResultsBySheet.isEmpty
}

var correctionCount: Int {
    userOverrides.count + forgottenOverrideKeys.count
}

var canUndoOverride: Bool {
    !overrideHistory.isEmpty
}

var manualOverridePositionsForCurrentSheet: Set<CellPosition> {
    guard let currentSheet else { return [] }
    return Set(
        userOverrides
        .filter { $0.sheetName == currentSheet }
        .map { CellPosition(row: $0.rowIndex, col: $0.colIndex) }
    )
}

var participatingFileCount: Int {
    validationReport?.includedFiles.count ?? 0
}

var blockedFileCount: Int {
    validationReport?.blockedFiles.count ?? 0
}

var warningFileCount: Int {
    validationReport?.warningFiles.count ?? 0
}

var sheetOverviewItems: [SheetOverviewItem] {
    guard let validationReport else { return [] }
    return WorkspaceDiagnostics.buildSheetOverview(
        report: validationReport,
        anomalyItems: allAnomalies
    )
}

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)
}

var toolbarPresentation: ToolbarPresentation {
    WorkspaceToolbar.buildPresentation(
        selectedFileCount: selectedFilePaths.count,
        canExport: canExport
    )
}

var selectedCellReference: String? {
    guard let selectedCell else { return nil }
    return WorkspaceDiagnostics.cellReference(row: selectedCell.row, col: selectedCell.col)
}

var currentSheet: String? {
    guard case .mergeable(let sheetName)? = selectedSheetSelection else { return nil }
    return sheetName
}

var selectedSheetName: String? {
    selectedSheetSelection?.sheetName
}

var selectedSheetStructureStatus: String {
    switch selectedSheetSelection {
    case .mergeable:
        return "可合并"
    case .skipped:
        return "已跳过"
    case .none:
        return validationReport?.readiness == .ready ? "可合并" : "-"
    }
}

var selectedSkippedSheetConsensus: SkippedSheetConsensus? {
    guard case .skipped(let sheetName)? = selectedSheetSelection,
        let validationReport else {
        return nil
    }
    return WorkspaceDiagnostics.buildSkippedSheetConsensus(
        report: validationReport,
        sheetName: sheetName
    )
}

var selectedMergedCell: MergedCell? {
    guard let selectedCell,
        let result = mergedResult,
        selectedCell.row < result.rows.count,
        selectedCell.col < result.rows[selectedCell.row].count else {
        return nil
    }
    return result.rows[selectedCell.row][selectedCell.col]
}

func correctionState(for position: CellPosition, cell: MergedCell? = nil) -> CellCorrectionState {
    if manualOverridePositionsForCurrentSheet.contains(position) {
        return .manual
    }
}

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if let cell {
    return cell.isOverridden ? .rule : .none
}

guard let result = mergedResult,
    position.row < result.rows.count,
    position.col < result.rows[position.row].count else {
    return .none
}

return result.rows[position.row][position.col].isOverridden ? .rule : .none
}

var canRestoreSelectedCellAutomatic: Bool {
    guard let currentSheet,
        let selectedCell else {
        return false
    }

    return effectiveWorkspaceOverrides().contains {
        $0.sheetName == currentSheet &&
        $0.rowIndex == selectedCell.row &&
        $0.colIndex == selectedCell.col
    }
}

/// 更新合并结果
private func revalidateSelection() {
    guard !selectedFilePaths.isEmpty else {
        closeAllFiles()
        return
    }

    isLoading = true
    workspacePhase = .validating

    Task {
        let batch = await parser.parseFilesWithDiagnostics(at: selectedFilePaths)
        let outcome = validator.validate(files: batch.files, parseFailures: batch.failures)

        await MainActor.run {
            validationReport = outcome.report
            files = outcome.mergeableFiles
            availableSheets = outcome.report.commonSheetNames
            isLoading = false
        }

        guard outcome.report.readiness == .ready, !outcome.mergeableFiles.isEmpty else {
            await MainActor.run {
                workspacePhase = .blocked
                selectedSheetSelection = nil
                mergedResult = nil
                mergedResultsBySheet.removeAll()
                selectedCell = nil
                anomalyQueue.removeAll()
                allAnomalies.removeAll()
                matchedSchema = nil
                workspaceRuleState = .none
            }
        }
    }
}

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return
    }

    await MainActor.run {
        workspacePhase = .ready
        selectedSheetSelection = normalizedSelectedSheetSelection(report: outcome.report)
    }
    await refreshWorkspaceResults()
}

}

private func refreshWorkspaceResults() async {
    guard workspacePhase == .ready, !files.isEmpty else { return }

    let ruleMatchResult = await smartMerger.prepareWorkspaceSchema(
        files: files,
        sheetNames: availableSheets
    )
    let workspaceSchema: MergeSchema?
    switch ruleMatchResult {
    case .exact(let schema):
        workspaceSchema = schema
    case .ambiguous, .similar, .none:
        workspaceSchema = nil
    }

    syncWorkspaceMemoryBase(with: workspaceSchema)
    let effectiveOverrides = effectiveWorkspaceOverrides()
    var resultsBySheet: [String: MergedResult] = [:]
    for sheetName in availableSheets {
        let result = await smartMerger.merge(
            files: files,
            sheetName: sheetName,
            applying: nil
        )
        let finalResult = smartMerger.applyOverrides(to: result, overrides: effectiveOverrides)
        resultsBySheet[sheetName] = finalResult
    }

    await MainActor.run {
        mergedResultsBySheet = resultsBySheet
        syncSelectedSheetPresentation()
        matchedSchema = workspaceSchema
        workspaceRuleState = Self.ruleState(from: ruleMatchResult)
        refreshSelectionAndAnomalies()
    }
}

}

/// 应用单元格类型覆盖
func applyCellOverride(row: Int, col: Int, type: CellOverrideType) {
    guard let currentSheet else { return }
    recordOverrideSnapshot()
    userOverrides.removeAll {
        $0.sheetName == currentSheet &&
        $0.rowIndex == row &&
        $0.colIndex == col
    }
    forgottenOverrideKeys.remove(
        StoredOverrideKey(sheetName: currentSheet, rowIndex: row, colIndex: col)
    )
}

```

```

let override = CellTypeOverride(
    sheetName: currentSheet,
    rowIndex: row,
    colIndex: col,
    cellType: type,
    userNote: nil
)
userOverrides.append(override)

Task {
    await persistAdjustmentMemoryAndRefresh()
}

}

/// 批量应用类型覆盖
func applyBulkOverride(positions: [CellPosition], type: CellOverrideType) {
    guard let currentSheet, !positions.isEmpty else { return }

    recordOverrideSnapshot()
    for pos in positions {
        userOverrides.removeAll {
            $0.sheetName == currentSheet &&
            $0.rowIndex == pos.row &&
            $0.colIndex == pos.col
        }
        forgottenOverrideKeys.remove(
            StoredOverrideKey(sheetName: currentSheet, rowIndex: pos.row, colIndex: pos.col)
        )
    }

    for pos in positions {
        let override = CellTypeOverride(
            sheetName: currentSheet,
            rowIndex: pos.row,
            colIndex: pos.col,
            cellType: type,
            userNote: nil
        )
        userOverrides.append(override)
    }

    Task {
        await persistAdjustmentMemoryAndRefresh()
    }
}

/// 清除所有用户覆盖
func clearOverrides() {
    guard correctionCount > 0 else { return }
    recordOverrideSnapshot()
    userOverrides.removeAll()
    forgottenOverrideKeys.removeAll()
    Task {
        await persistAdjustmentMemoryAndRefresh()
    }
}

/// 撤销上一步覆盖操作
func undoLastOverride() {

```



```

guard let lastOperation = overrideHistory.popLast() else { return }

switch lastOperation {
case .snapshot(let previousOverrides, let previousForgottenKeys):
    userOverrides = previousOverrides
    forgottenOverrideKeys = previousForgottenKeys
}

Task {
    await persistAdjustmentMemoryAndRefresh()
}

}

/// 显示规则管理器
func showSchemaManagerWindow() {
    showSchemaManager = true
}

func saveCurrentAdjustmentMemory() {
    guard canManageAdjustmentMemory else {
        WorkspaceDialogPresenter.runAlert(
            title: LocaleManager.loc("保存当前调整记忆"),
            message: LocaleManager.loc("当前没有可保存调整记忆的同构工作区。"),
            style: .informational
        )
        return
    }

    Task {
        await persistAdjustmentMemoryAndRefresh()
    }
}

func exportCurrentAdjustmentMemory() {
    guard let schema = currentAdjustmentMemory else { return }

    Task {
        do {
            let data = try await smartMerger.exportSchema(id: schema.id)
            let url = await MainActor.run {
                let panel = NSSavePanel()
                let fallbackName = schema.name.trimmingCharacters(in: .whitespacesAndNewLines).isEmpty
                    ? "adjustment-memory"
                    : schema.name
                panel.nameFieldStringValue = "\(fallbackName).json"
                panel.allowedContentTypes = [.json]
                return WorkspaceDialogPresenter.runModal(panel) == .OK ? panel.url : nil
            }
            if let url {
                try data.write(to: url)
            }
        } catch {
            await MainActor.run {
                errorMessage = "导出调整记忆失败: \(error.localizedDescription)"
                showError = true
            }
        }
    }
}
}

```

```

func showImportAdjustmentMemoryDialog() {
    guard canManageAdjustmentMemory else {
        WorkspaceDialogPresenter.runAlert(
            title: LocaleManager.loc("导入"),
            message: LocaleManager.loc("当前没有可绑定调整记忆的同构工作区。"),
            style: .informational
        )
        return
    }

    let panel = NSOpenPanel()
    panel.allowsMultipleSelection = false
    panel.canChooseDirectories = false
    panel.canChooseFiles = true
    panel.allowedContentTypes = [UTType.json]

    guard WorkspaceDialogPresenter.runModal(panel) == .OK, let url = panel.urls.first else { return }
    Task {
        await importCurrentAdjustmentMemory(from: url)
    }
}

func clearCurrentAdjustmentMemory() {
    guard let schemaID = workspaceActiveSchemaID ?? workspaceBaseSchemaID ?? matchedSchema?.id else { return }

    let alert = NSAlert()
    alert.alertStyle = .warning
    alert.messageText = LocaleManager.loc("清除当前调整记忆")
    alert.informativeText = LocaleManager.loc("删除当前调整记忆后，当前同构工作区将恢复自动判断。确定清除？")
    alert.addButton(withTitle: LocaleManager.loc("清除"))
    alert.addButton(withTitle: LocaleManager.loc("取消"))
    guard WorkspaceDialogPresenter.runAlert(alert) == .alertFirstButtonReturn else { return }

    Task {
        do {
            try await smartMerger.deleteSchema(id: schemaID)
            await MainActor.run {
                workspaceActiveSchemaID = nil
                workspaceBaseSchemaID = nil
                workspaceBaseOverrides.removeAll()
                userOverrides.removeAll()
                forgottenOverrideKeys.removeAll()
                overrideHistory.removeAll()
                matchedSchema = nil
                workspaceRuleState = .none
            }
            await refreshWorkspaceResults()
        } catch {
            await MainActor.run {
                errorMessage = "清除调整记忆失败: \(error.localizedDescription)"
                showError = true
            }
        }
    }
}

func selectCell(_ position: CellPosition?) {
    selectedCell = position
    refreshSelectionAndAnomalies()
}

```

```

func jumpToNextAnomaly() {
    jumpAcrossAnomalies(step: 1)
}

func jumpToPreviousAnomaly() {
    jumpAcrossAnomalies(step: -1)
}

private func jumpAcrossAnomalies(step: Int) {
    guard !allAnomalies.isEmpty else { return }

    let currentID = currentSheet.flatMap { sheetName in
        selectedCell.map { "\(sheetName)|\(WorkspaceDiagnostics.cellReference(row: $0.row, col: $0.col))" }
    }

    let currentIndex = currentID.flatMap { id in
        allAnomalies.firstIndex(where: { $0.id == id })
    } ?? (step > 0 ? -1 : 0)

    let nextIndex = (currentIndex + step + allAnomalies.count) % allAnomalies.count
    let target = allAnomalies[nextIndex]
    selectedSheetSelection = .mergeable(target.sheetName)
    syncSelectedSheetPresentation()
    selectedCell = target.position
    refreshSelectionAndAnomalies()
}

/// 导出当前结果
func exportResult() {
    guard canExport,
        let templatePath = validationReport?.templateFile?.filepath else { return }

    let panel = NSSavePanel()
    let exportName = ExportNaming.suggestedWorkbookName(
        from: validationReport?.files.map(\.filename) ??
            selectedFilePaths.map { URL(fileURLWithPath: $0).lastPathComponent }
    )
    panel.nameFieldStringValue = exportName
    panel.allowedContentTypes = [UTType(filenameExtension: "xlsx")!]

    if WorkspaceDialogPresenter.runModal(panel) == .OK, let url = panel.url {
        let results = availableSheets.compactMap { mergedResultsBySheet[$0] }
        let exporter = TemplateWorkbookExporter()
        Task {
            do {
                try exporter.exportWorkbook(
                    templatePath: templatePath,
                    results: results,
                    to: url.path
                )
            } catch {
                await MainActor.run {
                    errorMessage = "导出失败: \(error.localizedDescription)"
                    showError = true
                }
            }
        }
    }
}

```

```

private fun refreshSelectionAndAnomalies() {
    allAnomalies = availableSheets.flatMap { sheetName in
        mergedResultsBySheet[sheetName].map(WorkspaceDiagnostics.buildAnomalyQueue(for:)) ?? []
    }

    guard let currentSheet else {
        anomalyQueue = []
        selectedCell = nil
        return
    }

    anomalyQueue = mergedResultsBySheet[currentSheet].map(WorkspaceDiagnostics.buildAnomalyQueue(for:)) ?? []

    guard let result = mergedResult else {
        selectedCell = nil
        return
    }

    if let selectedCell,
        selectedCell.row < result.rows.count,
        selectedCell.col < result.rows[selectedCell.row].count {
        return
    }

    if let firstAnomaly = anomalyQueue.first {
        selectedCell = firstAnomaly.position
    } else if !result.rows.isEmpty, !result.rows[0].isEmpty {
        selectedCell = CellPosition(row: 0, col: 0)
    } else {
        selectedCell = nil
    }
}

private fun normalizedSelectedSheetSelection(report: WorkbookValidationReport?) -> WorkspaceSheetSelection? {
    if let selectedSheetSelection {
        switch selectedSheetSelection {
            case .mergeable(let sheetName) where availableSheets.contains(sheetName):
                return .mergeable(sheetName)
            case .skipped(let sheetName) where report?.skippedSheetNames.contains(sheetName) == true:
                return .skipped(sheetName)
            default:
                break
        }
    }

    if let firstMergeableSheet = availableSheets.first {
        return .mergeable(firstMergeableSheet)
    }

    if let firstSkippedSheet = report?.skippedSheetNames.first {
        return .skipped(firstSkippedSheet)
    }

    return nil
}

private fun syncSelectedSheetPresentation() {
    let normalizedSelection = normalizedSelectedSheetSelection(report: validationReport)
    if normalizedSelection != selectedSheetSelection {

```

```

selectedSheetSelection = normalizedSelection
}

switch normalizedSelection {
case .mergeable(let sheetName):
    mergedResult = mergedResultsBySheet[sheetName]
case .skipped:
    mergedResult = nil
case .none:
    mergedResult = nil
}
}

func restoreAutomaticDecisionForSelectedCell() {
    guard let currentSheet,
          let selectedCell,
          canRestoreSelectedCellAutomatic else {
        return
    }

    recordOverrideSnapshot()
    userOverrides.removeAll {
        $0.sheetName == currentSheet &&
        $0.rowIndex == selectedCell.row &&
        $0.colIndex == selectedCell.col
    }

    let key = StoredOverrideKey(sheetName: currentSheet, rowIndex: selectedCell.row, colIndex: selectedCell.col)
    if workspaceBaseOverrides.contains(where: {
        StoredOverrideKey(override: $0) == key
    }) {
        forgottenOverrideKeys.insert(key)
    } else {
        forgottenOverrideKeys.remove(key)
    }

    Task {
        await persistAdjustmentMemoryAndRefresh()
    }
}

private static func deduplicatedPaths(_ paths: [String]) -> [String] {
    var seen: Set<String> = []
    var result: [String] = []
    for path in paths where !seen.contains(path) {
        seen.insert(path)
        result.append(path)
    }
    return result
}

private static func ruleState(from matchResult: SchemaMatchResult) -> WorkspaceRuleState {
    switch matchResult {
    case .exact(let schema):
        return .applied(name: schema.name, correctionCount: schema.cellOverrides.count)
    case .ambiguous(let candidates):
        return .ambiguous(count: candidates.count)
    case .similar(let candidates):
        return .similar(count: candidates.count)
    case .none:

```

```

return .none
    }
}

private static func mergedRuleOverrides(
    existing: [CellTypeOverride],
    updates: [CellTypeOverride]
) -> [CellTypeOverride] {
    var merged = existing

    for update in updates {
        merged.removeAll {
            $0.sheetName == update.sheetName &&
            $0.rowIndex == update.rowIndex &&
            $0.colIndex == update.colIndex
        }
        merged.append(update)
    }

    return merged
}

private func recordOverrideSnapshot() {
    overrideHistory.append(
        .snapshot(
            previousOverrides: userOverrides,
            previousForgottenKeys: forgottenOverrideKeys
        )
    )
}

private func syncWorkspaceMemoryBase(with workspaceSchema: MergeSchema?) {
    guard userOverrides.isEmpty, forgottenOverrideKeys.isEmpty else { return }
    workspaceBaseOverrides = workspaceSchema?.cellOverrides ?? []
    workspaceBaseSchemaID = workspaceSchema?.id
    workspaceActiveSchemaID = workspaceSchema?.id
}

private func effectiveWorkspaceOverrides() -> [CellTypeOverride] {
    let rememberedOverrides = workspaceBaseOverrides.filter {
        !forgottenOverrideKeys.contains(StoredOverrideKey(override: $0))
    }
    return Self.mergedRuleOverrides(existing: rememberedOverrides, updates: userOverrides)
}

private func persistAdjustmentMemoryAndRefresh() async {
    do {
        try await persistAdjustmentMemory()
    } catch {
        errorMessage = "记住调整失败: \(error.localizedDescription)"
        showError = true
    }
    await refreshWorkspaceResults()
}

private func persistAdjustmentMemory() async throws {
    guard workspacePhase == .ready, !files.isEmpty, !availableSheets.isEmpty else { return }

    let fingerprint = FingerprintGenerator.generateWorkspaceLegacyFingerprint(
        from: files,

```

```

sheetNames: availableSheets
)
let workbookFingerprint = FingerprintGenerator.generateWorkbook(
    from: files,
    sheetNames: availableSheets
)
let effectiveOverrides = effectiveWorkspaceOverrides()
let schemaID = workspaceActiveSchemaID ?? workspaceBaseSchemaID

if effectiveOverrides.isEmpty {
    if let schemaID {
        if workspaceBaseSchemaID == nil {
            try await smartMerger.deleteSchema(id: schemaID)
            workspaceActiveSchemaID = nil
        } else {
            _ = try await smartMerger.updateSchema(
                id: schemaID,
                overrides: [],
                fingerprint: fingerprint,
                workbookFingerprint: workbookFingerprint
            )
            workspaceActiveSchemaID = schemaID
        }
    }
    return
}

if let schemaID {
    _ = try await smartMerger.updateSchema(
        id: schemaID,
        overrides: effectiveOverrides,
        fingerprint: fingerprint,
        workbookFingerprint: workbookFingerprint
    )
    workspaceActiveSchemaID = schemaID
} else {
    let schema = try await smartMerger.createSchema(
        name: suggestedAdjustmentMemoryName(),
        fingerprint: fingerprint,
        workbookFingerprint: workbookFingerprint,
        overrides: effectiveOverrides
    )
    workspaceActiveSchemaID = schema.id
}
}

private func importCurrentAdjustmentMemory(from url: URL) async {
    do {
        let data = try Data(contentsOf: url)
        let schema = try await smartMerger.importSchema(
            data: data,
            forCurrentWorkspaceFiles: files,
            sheetNames: availableSheets
        )
        await MainActor.run {
            workspaceBaseOverrides = schema.cellOverrides
            workspaceBaseSchemaID = schema.id
            workspaceActiveSchemaID = schema.id
            userOverrides.removeAll()
            forgottenOverrideKeys.removeAll()
        }
    }
}

```

```

overrideHistory.removeAll()
    }
    await refreshWorkspaceResults()
} catch {
    await MainActor.run {
        errorMessage = "导入调整记忆失败: \"(error.localizedDescription)\""
        showError = true
    }
}
}

private func suggestedAdjustmentMemoryName() -> String {
    let exportName = ExportNaming.suggestedWorkbookName(
        from: validationReport?.files.map(\.filename) ??
            selectedFilePaths.map { URL(fileURLWithPath: $0).lastPathComponent }
    )

    if exportName.hasSuffix("_汇总") {
        return String(exportName.dropLast("_汇总".count)) + "_调整记忆"
    }
    if exportName.hasSuffix("汇总") {
        return String(exportName.dropLast("汇总".count)) + "调整记忆"
    }
    return "\(exportName)_调整记忆"
}
}

// MARK: - 覆盖操作历史

private enum OverrideOperation {
    case snapshot(previousOverrides: [CellTypeOverride], previousForgottenKeys: Set<StoredOverrideKey>)
}

private struct StoredOverrideKey: Hashable {
    let sheetName: String
    let rowIndex: Int
    let colIndex: Int

    init(sheetName: String, rowIndex: Int, colIndex: Int) {
        self.sheetName = sheetName
        self.rowIndex = rowIndex
        self.colIndex = colIndex
    }

    init(override: CellTypeOverride) {
        self.sheetName = override.sheetName ?? ""
        self.rowIndex = override.rowIndex
        self.colIndex = override.colIndex
    }
}

// File: Sources/xlsOneUI/ContentView.swift
import SwiftUI
import xlsOneCore
import xlsOneLicense

struct ContentView: View {
    @EnvironmentObject var viewModel: AppViewModel
    @EnvironmentObject var licenseManager: LicenseManager
    @ObservedObject private var localeManager = LocaleManager.shared

```



```

@State private var isDropTargeted = false
@State private var showRestartToast = false
@State private var toastLanguage: LocaleManager.AppLanguage = .system
@State private var toastDismissTask: Task<Void, Never>?

var body: some View {
    VStack(spacing: 0) {
        toolbar
        Divider()

        switch viewModel.workspacePhase {
        case .idle:
            emptyView
        case .validating:
            loadingView
        case .blocked:
            blockedView
        case .ready:
            readyWorkspace
        }
    }
    .id(localeManager.currentLanguage.rawValue)
    .onChange(of: viewModel.showError) { isPresented in
        guard isPresented else { return }
        showErrorAlert()
    }
    .background {
        CenteredDialogWindow(
            isPresented: $viewModel.showSchemaManager,
            titleKey: "当前调整记忆",
            size: NSSize(width: 600, height: 400)
        ) {
            SchemaManagerView()
                .id(localeManager.currentLanguage.rawValue)
        }
        .frame(width: 0, height: 0)
    }
    .background {
        CenteredDialogWindow(
            isPresented: licenseActivationSheet,
            titleKey: "激活 表表归一",
            size: NSSize(width: 420, height: 520)
        ) {
            LicenseActivationView()
                .id(localeManager.currentLanguage.rawValue)
        }
        .frame(width: 0, height: 0)
    }
    .onDrop(of: [.fileURL], delegate: DropDelegateView(viewModel: viewModel, isTargeted: $isDropTargeted))
    .overlay {
        if showRestartToast {
            RestartToastView(language: toastLanguage)
                .transition(.opacity)
        }
    }
    .animation(.easeInOut(duration: 0.35), value: showRestartToast)
    .onReceive(NotificationCenter.default.publisher(for: .showRestartToast)) { notification in
        if let lang = notification.object as? LocaleManager.AppLanguage {
            toastLanguage = lang
        }
    }
}

```

```

toastDismissTask?.cancel()
    withAnimation {
        showRestartToast = true
    }
    toastDismissTask = Task { @MainActor in
        try? await Task.sleep(nanoseconds: 4_500_000_000)
        guard !Task.isCancelled else { return }
        withAnimation {
            showRestartToast = false
        }
    }
}
.task {
    await licenseManager.verifyOnLaunch()
}
}

private var licenseActivationSheet: Binding<Bool> {
    Binding(
        get: {
            if licenseManager.isAppStoreDistribution { return false }
            let state = licenseManager.licenseState
            if state == .activated { return false }
            if licenseManager.showActivationSheet {
                return true
            }
            if case .trial = state { return false }
            return state == .unactivated || state == .expired
        },
        set: { newValue in
            if !newValue {
                licenseManager.showActivationSheet = false
            }
        }
    )
}

private func showErrorAlert() {
    WorkspaceDialogPresenter.runAlert(
        title: LocaleManager.loc("错误"),
        message: viewModel.errorMessage ?? LocaleManager.loc("未知错误"),
        style: .warning
    )
    viewModel.showError = false
}

private var toolbar: some View {
    let presentation = viewModel.toolbarPresentation
    let hasWorkspace = !viewModel.selectedFilePaths.isEmpty

    return HStack(spacing: 10) {
        if hasWorkspace {
            HStack(spacing: 2) {
                toolbarUtilityButton(label: LocLabel("追加", systemImage: "plus"), help: LocaleManager.loc("向当前批次追加文件")) {
                    viewModel.showAddFileDialog()
                }
                .disabled(!presentation.appendEnabled)

                toolbarStripDivider
            }
        }
    }
}

```

```

toolbarUtilityButton(label: LocLabel("刷新", systemImage: "arrow.clockwise"), help: LocaleManager.loc("重新读取当前文件并刷新汇总结果")) {
    viewModel.reloadFiles()
}
.disabled(viewModel.selectedFilePaths.isEmpty)

toolbarUtilityButton(label: LocLabel("清空", systemImage: "xmark"), help: LocaleManager.loc("清空当前工作区，不影响原始 Excel 文件"))
{
    viewModel.closeAllFiles()
}
.disabled(viewModel.selectedFilePaths.isEmpty)
}
.padding(4)
.background(Color(NSColor.controlBackgroundColor))
.overlay(
    RoundedRectangle(cornerRadius: 12, style: .continuous)
        .stroke(Color.secondary.opacity(0.12), lineWidth: 1)
)
.clipShape(RoundedRectangle(cornerRadius: 12, style: .continuous))
}

Spacer()

if hasWorkspace {
    toolbarPrimaryButton(
        label: LocLabel("导出 XLSX", systemImage: "square.and.arrow.up"),
        prominent: presentation.exportIsProminent,
        help: viewModel.canExport ? LocaleManager.loc("导出同构汇总 Excel") : LocaleManager.loc("当前没有可导出的汇总结果")
    ) {
        viewModel.exportResult()
    }
    .disabled(!viewModel.canExport)
}

}
.padding(.horizontal)
.padding(.vertical, 10)
.background(Color(NSColor.windowBackgroundColor))
}

private var toolbarStripDivider: some View {
    Rectangle()
        .fill(Color.secondary.opacity(0.10))
        .frame(width: 1, height: 18)
        .padding(.horizontal, 2)
}

private func toolbarPrimaryButton(
    label: some View,
    prominent: Bool,
    help: String,
    action: @escaping () -> Void
) -> some View {
    Button(action: action) {
        label
    }
    .buttonStyle(WorkspaceChromePrimaryButtonStyle(prominent: prominent))
    .help(help)
}

private func toolbarUtilityButton(
    label: some View,
    help: String,

```

```

action: @escaping () -> Void
) -> some View {
    Button(action: action) {
        label
    }
    .buttonStyle(WorkspaceChromeUtilityButtonStyle())
    .help(help)
}

private var emptyView: some View {
    ZStack {
        LinearGradient(
            colors: [
                Color(NSColor.controlBackgroundColor),
                Color.white.opacity(0.92)
            ],
            startPoint: .top,
            endPoint: .bottom
        )

        RoundedRectangle(cornerRadius: 28, style: .continuous)
            .fill(Color.white.opacity(isDropTargeted ? 0.94 : 0.72))
            .frame(maxWidth: 560, maxHeight: 360)
            .overlay {
                EmptyWorkspaceBackdrop()
                .clipShape(RoundedRectangle(cornerRadius: 28, style: .continuous))
            }
            .overlay {
                RoundedRectangle(cornerRadius: 28, style: .continuous)
                .strokeBorder(dropZoneBorderColor, lineWidth: isDropTargeted ? 2 : 1)
            }
            .shadow(color: Color.black.opacity(isDropTargeted ? 0.08 : 0.04), radius: 24, y: 12)
            .overlay {
                VStack(spacing: 18) {
                    EmptyWorkspaceArtwork(isHighlighted: isDropTargeted)

                    VStack(spacing: 8) {
                        Text(isDropTargeted ? LocaleManager.loc("松手即可导入") : LocaleManager.loc("拖入 Excel 文件"))
                            .font(.system(size: 28, weight: .semibold))
                            .foregroundColor(.primary)

                        Text(isDropTargeted ? LocaleManager.loc("支持多个 .xlsx / .xls") : LocaleManager.loc("支持多个 .xlsx / .xls"))
                            .font(.subheadline)
                            .foregroundColor(.secondary)
                    }
                    .multilineTextAlignment(.center)

                    Button {
                        viewModel.showOpenFileDialog()
                    } label: {
                        Label(LocaleManager.loc("选择文件"), systemImage: "folder.badge.plus")
                    }
                    .buttonStyle(WorkspaceChromePrimaryButtonStyle(prominent: true))
                    .controlSize(.large)
                }
                .padding(.horizontal, 40)
                .padding(.vertical, 36)
            }
            .padding(40)
    }
}

```

```

.frame(maxWidth: .infinity, maxHeight: .infinity)
    .animation(.easeInOut(duration: 0.18), value: isDropTargeted)
}

private var dropZoneBorderColor: Color {
    isDropTargeted
        ? Color.accentColor.opacity(0.45)
        : Color.secondary.opacity(0.14)
}

private var loadingView: some View {
    VStack(spacing: 16) {
        ProgressView()
            .scaleEffect(1.4)
        Text(LocaleManager.loc("正在校验工作簿结构并准备汇总工作台..."))
            .foregroundColor(.secondary)
        Text(String(format: LocaleManager.loc("已选 %lld 个文件"), viewModel.selectedFilePaths.count))
            .font(.caption)
            .foregroundColor(.secondary)
    }
    .frame(maxWidth: .infinity, maxHeight: .infinity)
    .background(Color(NSColor.controlBackgroundColor))
}

private var blockedView: some View {
    ScrollView {
        VStack(alignment: .leading, spacing: 20) {
            VStack(alignment: .top, spacing: 12) {
                Image(systemName: "exclamationmark.triangle.fill")
                    .foregroundColor(.orange)
                    .font(.title2)
                VStack(alignment: .leading, spacing: 6) {
                    Text(LocaleManager.loc("没有可参与汇总的同构工作表"))
                        .font(.title3)
                        .fontWeight(.semibold)
                    Text(LocaleManager.loc("系统已忽略尾部空白行列后重试校验，但当前仍没有任何 sheet 能在所有文件间对齐。"))
                        .foregroundColor(.secondary)
                }
            }
        }

        if let report = viewModel.validationReport {
            validationSummary(report)
            if report.skippedSheetCount > 0 {
                skippedSheetList(report)
            }
            fileParticipationList(report.files)
        }
    }
    .padding(24)
    .frame(maxWidth: .infinity, alignment: .leading)
}
.background(Color(NSColor.controlBackgroundColor))
}

private func validationSummary(_ report: WorkbookValidationReport) -> some View {
    HStack(spacing: 12) {
        statCard(title: LocaleManager.loc("参与文件"), value: "\\(report.includedFiles.count)", tint: .green)
        statCard(title: LocaleManager.loc("阻断文件"), value: "\\(report.blockedFiles.count)", tint: .red)
        statCard(title: LocaleManager.loc("警告文件"), value: "\\(report.warningFiles.count)", tint: .orange)
        if report.skippedSheetCount > 0 {

```

```

statCard(title: LocaleManager.loc("跳过工作表"), value: "\\(report.skippedSheetCount)", tint: .orange)
    }
}

private func statCard(title: String, value: String, tint: Color) -> some View {
    VStack(alignment: .leading, spacing: 6) {
        Text(title)
            .font(.caption)
            .foregroundColor(.secondary)
        Text(value)
            .font(.title3)
            .fontWeight(.semibold)
            .foregroundColor(tint)
    }
    .padding()
    .frame(maxWidth: 140, alignment: .leading)
    .background(tint.opacity(0.1))
    .clipShape(RoundedRectangle(cornerRadius: 12))
}

private func fileParticipationList(_ reports: [FileValidationReport]) -> some View {
    VStack(alignment: .leading, spacing: 12) {
        Text(LocaleManager.loc("文件参与情况"))
            .font(.headline)

        ForEach(reports, id: \.filepath) { report in
            VStack(alignment: .leading, spacing: 10) {
                HStack(spacing: 10) {
                    statusDot(for: report.status)
                    Text(report.filename)
                        .fontWeight(.medium)
                    Spacer()
                    Text(report.statusLabel)
                        .font(.caption)
                        .foregroundColor(report.statusColor)
                }

                if !report.issues.isEmpty {
                    VStack(alignment: .leading, spacing: 6) {
                        ForEach(report.issues) { issue in
                            Text(issue.message)
                                .font(.caption)
                                .foregroundColor(issue.severity == .blocking ? .red : .orange)
                        }
                    }
                }
            }
        }
    }
    .padding()
    .background(Color.white)
    .clipShape(RoundedRectangle(cornerRadius: 12))
    .overlay(
        RoundedRectangle(cornerRadius: 12)
            .stroke(report.statusColor.opacity(0.2), lineWidth: 1)
    )
}

private func skippedSheetList(_ report: WorkbookValidationReport) -> some View {

```

```

case similar([SchemaMatchCandidate])

    /// 无匹配
    case none
}

/// 指纹相似度评分
public struct FingerprintSimilarity: Sendable {
    public let fingerprint: FileFingerprint
    public let score: Double // 0.0 - 1.0
    public let breakdown: SimilarityBreakdown

    public var isHighConfidence: Bool { score >= 0.9 }
    public var isMediumConfidence: Bool { score >= 0.7 }
}

/// 相似度评分细节
public struct SimilarityBreakdown: Sendable {
    public let sheetNameScore: Double
    public let dimensionScore: Double
    public let headerHashScore: Double
    public let sampleDataScore: Double
}

// File: Sources/xlsOneCore/Schema/SchemaMatcher.swift
import Foundation

/// Schema 匹配器 - 根据文件指纹匹配最合适的 Schema
public struct SchemaMatcher {

    /// 高置信度阈值
    public static let highConfidenceThreshold = 0.9

    /// 中置信度阈值
    public static let mediumConfidenceThreshold = 0.7

    /// 匹配 Schema
    /// - Parameters:
    ///   - fingerprint: 目标文件指纹
    ///   - schemas: 可用的 Schema 列表
    /// - Returns: 匹配结果
    public static func match(
        fingerprint: FileFingerprint,
        against schemas: [MergeSchema]
    ) -> SchemaMatchResult {
        guard !schemas.isEmpty else { return .none }

        // 计算每个 Schema 的相似度
        var similarities: [(schema: MergeSchema, score: Double, breakdown: SimilarityBreakdown)] = []

        for schema in schemas {
            let (score, breakdown) = calculateSimilarity(
                fingerprint,
                schema.fingerprint
            )
            similarities.append((schema, score, breakdown))
        }

        // 按相似度排序
        similarities.sort { $0.score > $1.score }

```

```

// 检查是否有高置信匹配；多个高置信命中时不静默任选
    let exactMatches = similarities.filter { $0.score >= highConfidenceThreshold }
    if exactMatches.count == 1, let exact = exactMatches.first {
        return .exact(exact.schema)
    }
    if exactMatches.count > 1 {
        return .ambiguous(exactMatches.map { SchemaMatchCandidate(schema: $0.schema, score: $0.score) })
    }

    // 返回相似匹配（中置信度及以上）
    let similarMatches = similarities.filter { $0.score >= mediumConfidenceThreshold }
    if !similarMatches.isEmpty {
        let results = similarMatches.map { SchemaMatchCandidate(schema: $0.schema, score: $0.score) }
        return .similar(results)
    }

    return .none
}

/// 匹配工作区级规则。一个工作区只能自动应用一套明确规则。
public static func match(
    workbookFingerprint: WorkbookRuleFingerprint,
    against schemas: [MergeSchema]
) -> SchemaMatchResult {
    guard !schemas.isEmpty else { return .none }

    var similarities: [(schema: MergeSchema, score: Double)] = []

    for schema in schemas {
        let score: Double
        if let savedWorkbookFingerprint = schema.workbookFingerprint {
            score = calculateWorkbookSimilarity(
                workbookFingerprint,
                savedWorkbookFingerprint
            )
        } else {
            score = calculateLegacySimilarity(
                workbookFingerprint: workbookFingerprint,
                schemaFingerprint: schema.fingerprint
            )
        }
        similarities.append((schema, score))
    }

    similarities.sort { lhs, rhs in
        if lhs.score != rhs.score {
            return lhs.score > rhs.score
        }
        return lhs.schema.updatedAt > rhs.schema.updatedAt
    }

    let exactMatches = similarities.filter { $0.score >= highConfidenceThreshold }
    if exactMatches.count == 1, let exact = exactMatches.first {
        return .exact(exact.schema)
    }
    if exactMatches.count > 1 {
        return .ambiguous(exactMatches.map { SchemaMatchCandidate(schema: $0.schema, score: $0.score) })
    }
}

```



```

let similarMatches = similarities.filter { $0.score >= mediumConfidenceThreshold }
    if !similarMatches.isEmpty {
        return .similar(similarMatches.map { SchemaMatchCandidate(schema: $0.schema, score: $0.score) })
    }

    return .none
}

/// 计算两个指纹的相似度
/// - Returns: (相似度分数 0-1, 评分细节)
public static func calculateSimilarity(
    _ fp1: FileFingerprint,
    _ fp2: FileFingerprint
) -> (score: Double, breakdown: SimilarityBreakdown) {
    var score = 0.0
    var totalWeight = 0.0

    // 1. 工作表名称匹配 (权重: 25%)
    totalWeight += 0.25
    let sheetNameScore = fp1.sheetName == fp2.sheetName ? 1.0 : 0.0
    score += 0.25 * sheetNameScore

    // 2. 行列维度匹配 (权重: 20%)
    // 允许 10% 的误差范围
    totalWeight += 0.20
    let rowRatio = ratioScore(fp1.rowCount, fp2.rowCount, tolerance: 0.1)
    let colRatio = ratioScore(fp1.colCount, fp2.colCount, tolerance: 0.1)
    let dimensionScore = (rowRatio + colRatio) / 2
    score += 0.20 * dimensionScore

    // 3. 表头哈希匹配 (权重: 35%)
    totalWeight += 0.35
    let headerScore = fp1.headerHash == fp2.headerHash ? 1.0 : 0.0
    score += 0.35 * headerScore

    // 4. 样本数据哈希匹配 (权重: 20%)
    totalWeight += 0.20
    let sampleScore = fp1.sampleDataHash == fp2.sampleDataHash ? 1.0 : 0.0
    score += 0.20 * sampleScore

    let breakdown = SimilarityBreakdown(
        sheetNameScore: sheetNameScore,
        dimensionScore: dimensionScore,
        headerHashScore: headerScore,
        sampleDataScore: sampleScore
    )

    // 归一化分数
    return (score / totalWeight, breakdown)
}

public static func calculateWorkbookSimilarity(
    _ fp1: WorkbookRuleFingerprint,
    _ fp2: WorkbookRuleFingerprint
) -> Double {
    guard !fp1.sheetFingerprints.isEmpty,
        fp1.sheetFingerprints.count == fp2.sheetFingerprints.count else {
        return 0
    }
}

```

```

var totalScore = 0.0
var sheetNamesAndDimensionsMatch = true

for (lhs, rhs) in zip(fp1.sheetFingerprints, fp2.sheetFingerprints) {
  if lhs.sheetName != rhs.sheetName ||
    lhs.rowCount != rhs.rowCount ||
    lhs.columnCount != rhs.columnCount {
    sheetNamesAndDimensionsMatch = false
  }

  var sheetScore = 0.0
  sheetScore += lhs.sheetName == rhs.sheetName ? 0.30 : 0
  sheetScore += (lhs.rowCount == rhs.rowCount && lhs.columnCount == rhs.columnCount) ? 0.30 : 0
  sheetScore += lhs.layoutHash == rhs.layoutHash ? 0.25 : 0
  sheetScore += lhs.formatHash == rhs.formatHash ? 0.15 : 0
  totalScore += sheetScore
}

let score = totalScore / Double(fp1.sheetFingerprints.count)
if sheetNamesAndDimensionsMatch {
  return max(score, highConfidenceThreshold)
}
return score
}

private static func calculateLegacySimilarity(
  workbookFingerprint: WorkbookRuleFingerprint,
  schemaFingerprint: FileFingerprint
) -> Double {
  guard let matchingSheet = workbookFingerprint.sheetFingerprints.first(where: {
    $0.sheetName == schemaFingerprint.sheetName
  }) else {
    return 0
  }

  let dimensionsMatch = matchingSheet.rowCount == schemaFingerprint.rowCount &&
    matchingSheet.columnCount == schemaFingerprint.colCount
  var score = 0.0
  score += matchingSheet.sheetName == schemaFingerprint.sheetName ? 0.30 : 0
  score += dimensionsMatch ? 0.30 : 0
  score += matchingSheet.layoutHash == schemaFingerprint.headerHash ? 0.25 : 0
  score += matchingSheet.formatHash == schemaFingerprint.sampleDataHash ? 0.15 : 0

  if schemaFingerprint.schemaVersion < 2,
    dimensionsMatch {
    let legacyFloor = workbookFingerprint.sheetFingerprints.count == 1
      ? highConfidenceThreshold
      : mediumConfidenceThreshold
    score = max(score, legacyFloor)
  }

  return score
}

/// 计算比例相似度
/// - Parameters:
///   - v1: 值1
///   - v2: 值2
///   - tolerance: 允许误差比例
/// - Returns: 相似度 0-1

```

```

private static func ratioScore(_ v1: Int, _ v2: Int, tolerance: Double) -> Double {
    if v1 == v2 { return 1.0 }
    if v1 == 0 || v2 == 0 { return 0.0 }

    let ratio = Double(min(v1, v2)) / Double(max(v1, v2))
    if ratio >= (1.0 - tolerance) {
        return ratio
    }
    return 0.0
}

/// 查找最佳匹配的 Schema (简化接口)
public static func findBestMatch(
    fingerprint: FileFingerprint,
    schemas: [MergeSchema]
) -> MergeSchema? {
    let result = match(fingerprint: fingerprint, against: schemas)

    switch result {
    case .exact(let schema):
        return schema
    case .ambiguous:
        return nil
    case .similar(let matches):
        return matches.first?.schema
    case .none:
        return nil
    }
}

}

// File: Sources/xlsOneCore/Schema/SchemaRepository.swift
import Foundation

/// Schema 存储仓库协议
public protocol SchemaRepositoryProtocol: Sendable {
    /// 加载所有 Schema
    func loadAllSchemas() async throws -> [MergeSchema]

    /// 保存 Schema
    func saveSchema(_ schema: MergeSchema) async throws

    /// 删除 Schema
    func deleteSchema(id: UUID) async throws

    /// 根据 ID 查找 Schema
    func findSchema(id: UUID) async throws -> MergeSchema?

    /// 根据指纹查找匹配的 Schema
    func findMatchingSchema(fingerprint: FileFingerprint) async throws -> SchemaMatchResult

    /// 根据工作区指纹查找匹配的 Schema
    func findMatchingSchema(workbookFingerprint: WorkbookRuleFingerprint) async throws -> SchemaMatchResult

    /// 更新 Schema 匹配计数
    func incrementMatchCount(id: UUID) async throws

    /// 导出 Schema
    func exportSchema(id: UUID) async throws -> Data

```

```

/// 导入 Schema
func importSchema(data: Data) async throws -> MergeSchema
}

/// 本地文件系统实现的 Schema 仓库
public actor SchemaRepository: SchemaRepositoryProtocol {

    /// 单例实例
    public static let shared = SchemaRepository()

    /// Schema 存储目录
    private let baseDirectory: URL

    /// Schema 索引文件名
    private let indexFileName = "index.json"

    /// 内存中的缓存
    private var cachedSchemas: [MergeSchema]?

    /// 初始化
    public init(baseDirectory: URL? = nil) {
        if let baseDir = baseDirectory {
            self.baseDirectory = baseDir
        } else {
            // 使用应用支持目录
            let appSupport = FileManager.default.urls(
                for: .applicationSupportDirectory,
                in: .userDomainMask
            ).first!
            self.baseDirectory = appSupport.appendingPathComponent("xlsOne/schemas", isDirectory: true)
        }
    }

    // MARK: - SchemaRepositoryProtocol

    public func loadAllSchemas() async throws -> [MergeSchema] {
        // 返回缓存（如果可用）
        if let cached = cachedSchemas {
            return cached
        }

        // 确保目录存在
        try ensureDirectoryExists()

        // 读取索引文件
        let indexURL = baseDirectory.appendingPathComponent(indexFileName)
        guard FileManager.default.fileExists(atPath: indexURL.path) else {
            cachedSchemas = []
            return []
        }

        let data = try Data(contentsOf: indexURL)
        let ids = try JSONDecoder().decode([UUID].self, from: data)

        // 加载每个 Schema
        var schemas: [MergeSchema] = []
        for id in ids {
            if let schema = try await loadSchema(id: id) {
                schemas.append(schema)
            }
        }
    }

```

```

}

// 按匹配次数排序（常用优先）
schemas.sort { $0.updatedAt > $1.updatedAt }

cachedSchemas = schemas
return schemas
}

public func saveSchema(_ schema: MergeSchema) async throws {
    try ensureDirectoryExists()

    // 保存 Schema 文件
    let schemaURL = schemaFileURL(id: schema.id)
    let data = try JSONEncoder().encode(schema)
    try data.write(to: schemaURL)

    // 更新索引
    var ids = try loadIndex()
    if !ids.contains(schema.id) {
        ids.append(schema.id)
        try saveIndex(ids)
    }

    // 更新缓存
    if cachedSchemas != nil {
        // 移除旧的（如果存在）
        cachedSchemas?.removeAll { $0.id == schema.id }
        // 添加新的
        cachedSchemas?.append(schema)
    }
}

public func deleteSchema(id: UUID) async throws {
    // 删除文件
    let schemaURL = schemaFileURL(id: id)
    if FileManager.default.fileExists(atPath: schemaURL.path) {
        try FileManager.default.removeItem(at: schemaURL)
    }

    // 更新索引
    var ids = try loadIndex()
    ids.removeAll { $0 == id }
    try saveIndex(ids)

    // 更新缓存
    cachedSchemas?.removeAll { $0.id == id }
}

public func findSchema(id: UUID) async throws -> MergeSchema? {
    // 先查缓存
    if let cached = cachedSchemas?.first(where: { $0.id == id }) {
        return cached
    }

    // 从文件加载
    return try await loadSchema(id: id)
}

public func findMatchingSchema(fingerprint: FileFingerprint) async throws -> SchemaMatchResult {

```

```

let schemas = try await loadAllSchemas()
    return SchemaMatcher.match(fingerprint: fingerprint, against: schemas)
}

public func findMatchingSchema(workbookFingerprint: WorkbookRuleFingerprint) async throws -> SchemaMatchResult {
    let schemas = try await loadAllSchemas()
    return SchemaMatcher.match(workbookFingerprint: workbookFingerprint, against: schemas)
}

public func incrementMatchCount(id: UUID) async throws {
    guard let schema = try await findSchema(id: id) else { return }

    // 创建更新后的 Schema
    let updatedSchema = MergeSchema(
        id: schema.id,
        name: schema.name,
        fingerprint: schema.fingerprint,
        workbookFingerprint: schema.workbookFingerprint,
        cellOverrides: schema.cellOverrides,
        createdAt: schema.createdAt,
        updatedAt: Date()
    )

    try await saveSchema(updatedSchema)
}

// MARK: - 导入/导出

/// 导出 Schema 为 JSON 数据
public func exportSchema(id: UUID) async throws -> Data {
    guard let schema = try await findSchema(id: id) else {
        throw SchemaRepositoryError.schemaNotFound
    }

    let wrapper = ExportedSchema(
        version: "1.0",
        exportedAt: Date(),
        schema: schema
    )

    return try JSONEncoder().encode(wrapper)
}

/// 从 JSON 数据导入 Schema
public func importSchema(data: Data) async throws -> MergeSchema {
    let newSchema = try Self.decodeImportableSchema(data: data)
    try await saveSchema(newSchema)
    return newSchema
}

/// 解析可导入的调整记忆，但不写入本地仓库。
public nonisolated static func decodeImportableSchema(data: Data) throws -> MergeSchema {
    let decoder = JSONDecoder()

    if let wrapper = try? decoder.decode(ExportedSchema.self, from: data) {
        return importedCopy(of: wrapper.schema)
    }

    let schema = try decoder.decode(MergeSchema.self, from: data)
    return importedCopy(of: schema)
}

```

```

}

// MARK: - 私有方法

private func ensureDirectoryExists() throws {
    if !FileManager.default.fileExists(atPath: baseDirectory.path) {
        try FileManager.default.createDirectory(
            at: baseDirectory,
            withIntermediateDirectories: true
        )
    }
}

private func schemaFileURL(id: UUID) -> URL {
    baseDirectory.appendingPathComponent("schema-\(id.uuidString).json")
}

private func loadSchema(id: UUID) async throws -> MergeSchema? {
    let url = schemaFileURL(id: id)
    guard FileManager.default.fileExists(atPath: url.path) else { return nil }

    let data = try Data(contentsOf: url)
    return try JSONDecoder().decode(MergeSchema.self, from: data)
}

private func loadIndex() throws -> [UUID] {
    let url = baseDirectory.appendingPathComponent(indexFileName)
    guard FileManager.default.fileExists(atPath: url.path) else { return [] }

    let data = try Data(contentsOf: url)
    return try JSONDecoder().decode([UUID].self, from: data)
}

private func saveIndex(_ ids: [UUID]) throws {
    let url = baseDirectory.appendingPathComponent(indexFileName)
    let data = try JSONEncoder().encode(ids)
    try data.write(to: url)
}

private nonisolated static func importedCopy(of schema: MergeSchema) -> MergeSchema {
    MergeSchema(
        id: UUID(),
        name: schema.name + " (导入)",
        fingerprint: schema.fingerprint,
        workbookFingerprint: schema.workbookFingerprint,
        cellOverrides: schema.cellOverrides
    )
}

/// 导出的 Schema 包装器
private struct ExportedSchema: Codable {
    let version: String
    let exportedAt: Date
    let schema: MergeSchema
}

/// 仓库错误
public enum SchemaRepositoryError: Error {
    case schemaNotFound

```

```

case invalidData
case saveFailed
}

// File: Sources/xlsOneCore/Schema/FingerprintGenerator.swift
import Foundation
import CryptoKit

/// 文件指纹生成器
public struct FingerprintGenerator {

    /// 从 ExcelFile 生成指纹
    /// - Parameters:
    ///   - file: Excel 文件
    ///   - fileNamePattern: 可选的文件名匹配模式
    /// - Returns: 文件指纹
    public static func generate(from file: ExcelFile, fileNamePattern: String? = nil) -> FileFingerprint {
        // 使用第一个工作表作为主要特征源
        guard let sheet = file.sheets.first else {
            return FileFingerprint(
                sheetName: "",
                rowCount: 0,
                colCount: 0,
                headerHash: "",
                sampleDataHash: "",
                fileNamePattern: fileNamePattern
            )
        }

        let headerHash = generateHeaderHash(sheet: sheet)
        let sampleDataHash = generateSampleDataHash(sheet: sheet)
        let maxCols = sheet.rows.map { $0.count }.max() ?? 0

        return FileFingerprint(
            sheetName: sheet.name,
            rowCount: sheet.rows.count,
            colCount: maxCols,
            headerHash: headerHash,
            sampleDataHash: sampleDataHash,
            fileNamePattern: fileNamePattern
        )
    }

    /// 从指定工作表生成兼容旧规则的单表指纹
    public static func generate(
        from file: ExcelFile,
        sheetName: String,
        fileNamePattern: String? = nil
    ) -> FileFingerprint {
        guard let sheet = file.sheets.first(where: { $0.name == sheetName }) else {
            return FileFingerprint(
                sheetName: sheetName,
                rowCount: 0,
                colCount: 0,
                headerHash: "",
                sampleDataHash: "",
                fileNamePattern: fileNamePattern
            )
        }
    }
}

```



```

let dimensions = effectiveDimensions(for: sheet)
    let sheetFingerprint = generateSheetRuleFingerprint(
        sheet: sheet,
        effectiveRowCount: dimensions.rows,
        effectiveColumnCount: dimensions.cols
    )

    return FileFingerprint(
        schemaVersion: 2,
        sheetName: sheet.name,
        rowCount: sheetFingerprint.rowCount,
        colCount: sheetFingerprint.columnCount,
        headerHash: sheetFingerprint.layoutHash,
        sampleDataHash: sheetFingerprint.formatHash,
        fileNamePattern: fileNamePattern
    )
}

/// 从一组同构 Excel 的可合并工作表生成工作区级规则指纹
public static func generateWorkbook(
    from files: [ExcelFile],
    sheetNames: [String]
) -> WorkbookRuleFingerprint {
    let fingerprints = sheetNames.map { sheetName in
        generateConsensusSheetFingerprint(from: files, sheetName: sheetName)
    }

    return WorkbookRuleFingerprint(sheetFingerprints: fingerprints)
}

/// 为新规则提供兼容旧字段的代表性指纹
public static func generateWorkspaceLegacyFingerprint(
    from files: [ExcelFile],
    sheetNames: [String]
) -> FileFingerprint {
    guard let firstSheetName = sheetNames.first,
        let file = files.first else {
        return FileFingerprint(
            schemaVersion: 2,
            sheetName: "",
            rowCount: 0,
            colCount: 0,
            headerHash: "",
            sampleDataHash: ""
        )
    }

    return generate(from: file, sheetName: firstSheetName)
}

/// 生成表头哈希
/// 结合工作表名称、第一行（列头）和第一列（行标签）
private static func generateHeaderHash(sheet: SheetData) -> String {
    var components: [String] = []

    // 添加工作表名称
    components.append(sheet.name)

    // 添加第一行内容（列头）- 最多前 10 列
    if let firstRow = sheet.rows.first {

```

```

let limit = min(firstRow.count, 10)
    for i in 0..

```

```

.replacingOccurrences(of: "\\s+", with: "", options: .regularExpression)
}

/// 计算字符串哈希 (使用 SHA256 的前 16 位)
private static func hash(_ text: String) -> String {
    let data = Data(text.utf8)
    let hash = SHA256.hash(data: data)
    // 取前 8 字节 (16 个十六进制字符) 作为哈希值
    return hash.prefix(8).map { String(format: "%02x", $0) }.joined()
}

private static func generateConsensusSheetFingerprint(
    from files: [ExcelFile],
    sheetName: String
) -> SheetRuleFingerprint {
    let sheets = files.compactMap { file in
        file.sheets.first(where: { $0.name == sheetName })
    }

    let dimensions = chooseDominantDimensions(
        from: sheets.map(effectiveDimensions)
    )

    guard let representativeSheet = sheets.first else {
        return SheetRuleFingerprint(
            sheetName: sheetName,
            rowCount: 0,
            columnCount: 0,
            layoutHash: "",
            formatHash: ""
        )
    }

    return generateSheetRuleFingerprint(
        sheet: representativeSheet,
        effectiveRowCount: dimensions.rows,
        effectiveColumnCount: dimensions.cols
    )
}

private static func generateSheetRuleFingerprint(
    sheet: SheetData,
    effectiveRowCount: Int,
    effectiveColumnCount: Int
) -> SheetRuleFingerprint {
    var layoutComponents: [String] = [sheet.name, "\\(effectiveRowCount)x\\(effectiveColumnCount)"]
    var formatComponents: [String] = []

    for rowIndex in 0..

```

```

columnCount: effectiveColumnCount,
    layoutHash: hash(layoutComponents.joined(separator: "|")),
    formatHash: hash(formatComponents.joined(separator: "|"))
)
}

private static func structuralToken(for cell: CellData?) -> String {
    guard let cell else { return "E" }
    let value = cell.value.trimmingCharacters(in: .whitespacesAndNewLines)
    guard !value.isEmpty else { return "E" }
    if cell.isDate { return "D" }
    if cell.numericValue != nil { return "N" }
    if value.range(of: #"\\p{Han}"#, options: .regularExpression) != nil { return "C" }
    if value.range(of: #"[A-Za-z]"#, options: .regularExpression) != nil { return "A" }
    return "T"
}

private static func effectiveDimensions(for sheet: SheetData) -> SheetDimensions {
    var lastNonEmptyRow = -1
    var lastNonEmptyColumn = -1

    for (rowIndex, row) in sheet.rows.enumerated() {
        for (columnIndex, cell) in row.enumerated()
        where !cell.value.trimmingCharacters(in: .whitespacesAndNewLines).isEmpty {
            lastNonEmptyRow = max(lastNonEmptyRow, rowIndex)
            lastNonEmptyColumn = max(lastNonEmptyColumn, columnIndex)
        }
    }

    return SheetDimensions(
        rows: max(lastNonEmptyRow + 1, 0),
        cols: max(lastNonEmptyColumn + 1, 0)
    )
}

private static func chooseDominantDimensions(from dimensions: [SheetDimensions]) -> SheetDimensions {
    guard !dimensions.isEmpty else { return SheetDimensions(rows: 0, cols: 0) }

    return Dictionary(grouping: dimensions.enumerated(), by: \.element)
        .values
        .max { lhs, rhs in
            if lhs.count != rhs.count {
                return lhs.count < rhs.count
            }

            let lhsDimensions = lhs[0].element
            let rhsDimensions = rhs[0].element
            if lhsDimensions != rhsDimensions {
                return lhsDimensions < rhsDimensions
            }

            let lhsFirstIndex = lhs.map(\.offset).min() ?? .max
            let rhsFirstIndex = rhs.map(\.offset).min() ?? .max
            return lhsFirstIndex > rhsFirstIndex
        }?
        .first?
        .element ?? SheetDimensions(rows: 0, cols: 0)
}
}

```

```

private struct SheetDimensions: Hashable, Comparable {
    let rows: Int
    let cols: Int

    static func < (lhs: SheetDimensions, rhs: SheetDimensions) -> Bool {
        if lhs.rows != rhs.rows {
            return lhs.rows < rhs.rows
        }
        return lhs.cols < rhs.cols
    }
}

// File: Sources/xlsOneLicense/DeviceFingerprint.swift
import Foundation
import CryptoKit

/// Generates a stable device fingerprint for license binding.
/// Uses a Keychain-stored UUID — survives reboots, survives Keychain reset
/// only if the user explicitly removes the entry.
enum DeviceFingerprint {

    static func generate() -> String {
        // Primary: Keychain-stored UUID (App Store compatible, no IOKit needed)
        if let existing = Keychain.load(key: "com.xlsone.deviceUUID") {
            return sha256(existing + ".xlsone.device")
        }

        // First launch: generate and persist a new UUID
        let uuid = UUID().uuidString
        _ = Keychain.save(key: "com.xlsone.deviceUUID", data: uuid)
        return sha256(uuid + ".xlsone.device")
    }

    // MARK: - Private

    private static func sha256(_ input: String) -> String {
        let data = Data(input.utf8)
        let hash = SHA256.hash(data: data)
        return hash.compactMap { String(format: "%02x", $0) }.joined()
    }
}

```

```

// File: Sources/xlsOneLicense/Keychain.swift
import Foundation
import Security

/// Minimal Keychain wrapper for storing license tokens and device identifiers.
enum Keychain {

    static func save(key: String, data: String) -> Bool {
        delete(key: key)

        let query: [String: Any] = [
            kSecClass as String: kSecClassGenericPassword,
            kSecAttrAccount as String: key,
            kSecAttrService as String: "com.xlsone.license",
            kSecValueData as String: Data(data.utf8),
            kSecAttrAccessible as String: kSecAttrAccessibleAfterFirstUnlock,
        ]
    }
}

```

```

let status = SecItemAdd(query as CFDictionary, nil)
    return status == errSecSuccess
}

static func load(key: String) -> String? {
    let query: [String: Any] = [
        kSecClass as String: kSecClassGenericPassword,
        kSecAttrAccount as String: key,
        kSecAttrService as String: "com.xlsone.license",
        kSecReturnData as String: true,
        kSecMatchLimit as String: kSecMatchLimitOne,
    ]

    var item: CTypeRef?
    let status = SecItemCopyMatching(query as CFDictionary, &item)
    guard status == errSecSuccess, let data = item as? Data else { return nil }

    return String(data: data, encoding: .utf8)
}

static func delete(key: String) {
    let query: [String: Any] = [
        kSecClass as String: kSecClassGenericPassword,
        kSecAttrAccount as String: key,
        kSecAttrService as String: "com.xlsone.license",
    ]
    SecItemDelete(query as CFDictionary)
}
}

// File: Sources/xlsOneLicense/LicenseManager.swift
import Foundation
import xlsOneCore
#if os(macOS)
import AppKit
#endif

// MARK: - License Manager

/// Manages software license activation, verification, and offline support.
/// Communicates with the xlsOne activation API for online validation.
public final class LicenseManager: ObservableObject {

    public static let shared = LicenseManager()
    public static var isAppStoreDistribution: Bool {
        Bundle.main.object(forKey: "XLSONEDistributionChannel") as? String == "app-store"
    }

    // MARK: - Published State

    @Published public private(set) var licenseState: LicenseState = .unactivated
    @Published public private(set) var plan: LicensePlan = .unknown
    @Published public private(set) var isVerifying = false
    @Published public var showActivationSheet = false

    // MARK: - Constants

    private enum API {
        /// Primary endpoint (Cloudflare Workers — global)
        static let primary = "https://api.xlsone.com"
    }

```

```

/// Fallback endpoint (Aliyun FC — China mainland)
static let fallback = "https://"
/// Request timeout
static let timeout: TimeInterval = 3.0
}

private enum Storage {
    static let tokenKey = "com.xlsone.license.token"
    static let deviceIDKey = "com.xlsone.license.deviceID"
    static let offlineLicenseKey = "com.xlsone.license.offline"
    static let trialStartKey = "com.xlsone.license.trialStart"
}

private static let trialDurationDays = 14

// MARK: - Initialization

private init() {
    if Self.isAppStoreDistribution {
        licenseState = .activated
        plan = .appStore
    } else {
        loadPersistedState()
    }
}

// MARK: - Public API

/// Start a free trial. Returns the remaining days.
public func startTrial() -> Int {
    if Self.isAppStoreDistribution {
        licenseState = .activated
        plan = .appStore
        return Int.max
    }

    let now = Date()
    UserDefaults.standard.set(now, forKey: Storage.trialStartKey)
    let remaining = Self.trialDurationDays
    licenseState = .trial(remainingDays: remaining)
    return remaining
}

/// Check trial status. Returns remaining days or -1 if trial expired/not started.
public func checkTrialStatus() -> Int {
    guard let start = UserDefaults.standard.object(forKey: Storage.trialStartKey) as? Date else {
        return -1
    }
    let elapsed = Calendar.current.dateComponents([.day], from: start, to: Date()).day ?? 0
    let remaining = Self.trialDurationDays - elapsed
    if remaining > 0 {
        return remaining
    }
    return -1
}

/// Import an offline license file (JWT token).
@discardableResult
@MainActor
public func importOfflineLicenseFile() -> ActivationResult? {

```

```

guard !Self.isAppStoreDistribution else {
    return .success(plan: .appStore, expiresAt: nil)
}

#if os(macOS)
    let panel = NSOpenPanel()
    panel.allowedContentTypes = [.data, .json]
    panel.allowsMultipleSelection = false
    panel.message = "选择授权文件"

    guard Self.runCenteredModal(panel) == .OK, let url = panel.url else { return nil }

    guard let token = try? String(contentsOf: url, encoding: .utf8).trimmingCharacters(in: .whitespacesAndNewlines),
        !token.isEmpty
    else {
        DispatchQueue.main.async {
            self.licenseState = .unactivated
        }
        return .failure(.invalidLicenseFile)
    }

    let deviceID = getOrCreateDeviceID()
    guard let payload = decodeTokenPayload(token) else {
        DispatchQueue.main.async {
            self.licenseState = .unactivated
        }
        return .failure(.invalidLicenseFile)
    }

    guard payload.dev == deviceID else {
        return .failure(.licenseDeviceMismatch)
    }

    guard isValidTokenLocally(payload, deviceID: deviceID) else {
        return .failure(.invalidLicenseFile)
    }

    let plan = LicensePlan(rawValue: payload.plan) ?? .unknown
    saveToken(token)
    UserDefaults.standard.set(token, forKey: Storage.offlineLicenseKey)
    DispatchQueue.main.async {
        self.licenseState = .activated
        self.plan = plan
    }
    return .success(plan: plan, expiresAt: nil)
#else
    return .failure(.serverError(LocaleManager.loc("当前平台不支持导入授权文件")))
#endif
}

/// Activate with an activation key. Returns the result.
public func activate(key: String) async -> ActivationResult {
    if Self.isAppStoreDistribution {
        await MainActor.run {
            self.licenseState = .activated
            self.plan = .appStore
        }
        return .success(plan: .appStore, expiresAt: nil)
    }
}

```



```

let deviceID = getOrCreateDeviceID()
    let normalizedKey = key.uppercased().trimmingCharacters(in: .whitespaces)

    guard KeyFormat.isValid(normalizedKey) else {
        return .failure(.invalidKeyFormat)
    }

    let body: [String: String] = [
        "key": normalizedKey,
        "device_id": deviceID,
        "device_name": Host.current().localizedName ?? "Unknown Mac"
    ]

    // Try primary endpoint first
    if let result = await postActivation(to: API.primary, body: body) {
        return result
    }

    // Fallback to China endpoint
    if let result = await postActivation(to: API.fallback, body: body) {
        return result
    }

    // Allow offline activation attempt from stored offline license
    if let offlineResult = tryOfflineActivation(key: normalizedKey, deviceID: deviceID) {
        return offlineResult
    }

    return .failure(.networkError)
}

/// Verify current license validity. Called on app launch.
public func verifyOnLaunch() async {
    if Self.isAppStoreDistribution {
        await MainActor.run {
            self.licenseState = .activated
            self.plan = .appStore
            self.showActivationSheet = false
        }
        return
    }

    await MainActor.run { isVerifying = true }
    defer { Task { @MainActor in isVerifying = false } }

    guard let token = loadToken() else {
        await setState(.unactivated)
        return
    }

    let deviceID = getOrCreateDeviceID()

    // Try online verification
    if let result = await postVerify(to: API.primary, token: token, deviceID: deviceID) {
        await handleVerifyResult(result)
        return
    }

    // Fallback endpoint
    if let result = await postVerify(to: API.fallback, token: token, deviceID: deviceID) {

```

```

await handleVerifyResult(result)
    return
}

// Offline validation — check token locally
if let payload = decodeTokenPayload(token), isTokenValidLocally(payload, deviceID: deviceID) {
    await setState(.activated)
    await setPlan(LicensePlan(rawValue: payload.plan) ?? .unknown)
    return
}

// Token invalid, try offline license file
if let offlineToken = loadOfflineLicense(), let payload = decodeTokenPayload(offlineToken),
    isTokenValidLocally(payload, deviceID: deviceID) {
    await setState(.activated)
    await setPlan(LicensePlan(rawValue: payload.plan) ?? .unknown)
    return
}

// Grace period: if token was valid recently (within 7 days), allow
if let lastValid = UserDefaults.standard.object(forKey: "com.xlsone.license.lastValid") as? Date,
    Date().timeIntervalSince(lastValid) < 7 * 24 * 60 * 60 {
    await setState(.gracePeriod)
    return
}

await setState(.expired)
}

/// Refresh subscription token. Called periodically.
public func refreshIfNeeded() async {
    guard !Self.isAppStoreDistribution else { return }

    guard case .activated = licenseState,
        let token = loadToken(),
        let payload = decodeTokenPayload(token),
        payload.exp > 0, // Lifetime licenses don't need refresh
        payload.exp < Int(Date().timeIntervalSince1970) + 7 * 24 * 60 * 60 // refresh if expiring within 7 days
    else { return }

    let deviceID = getOrCreateDeviceID()

    let body: [String: String] = ["token": token, "device_id": deviceID]

    if let data = await postJSON(to: "\(API.primary)/api/refresh", body: body),
        let result = try? JSONDecoder().decode(RefreshResponse.self, from: data),
        result.valid {
        if let token = result.token { saveToken(token) }
    }
}

/// Reset license (for user logout / key change)
public func reset() {
    guard !Self.isAppStoreDistribution else {
        licenseState = .activated
        plan = .appStore
        showActivationSheet = false
        return
    }
}

```

```

Keychain.delete(key: Storage.tokenKey)
    UserDefaults.standard.removeObject(forKey: Storage.offlineLicenseKey)
    licenseState = .unactivated
    plan = .unknown
}

// MARK: - Private — Network

private func postActivation(to base: String, body: [String: String]) async -> ActivationResult? {
    guard let data = await postJSON(to: "\\(base)/api/activate", body: body) else {
        return nil
    }

    guard let result = try? JSONDecoder().decode(ActivateResponse.self, from: data) else {
        if let error = try? JSONDecoder().decode(APIError.self, from: data) {
            return .failure(mapAPIError(error))
        }
        return nil
    }

    saveToken(result.license_token)

    let plan = LicensePlan(rawValue: result.plan) ?? .personalYearly
    Task { @MainActor in
        self.licenseState = .activated
        self.plan = plan
        UserDefaults.standard.set(Date(), forKey: "com.xlsone.license.lastValid")
    }

    return .success(plan: plan, expiresAt: result.expires_at)
}

private func postVerify(to base: String, token: String, deviceID: String) async -> VerifyResponse? {
    let body: [String: String] = ["token": token, "device_id": deviceID]
    guard let data = await postJSON(to: "\\(base)/api/verify", body: body) else {
        return nil
    }
    return try? JSONDecoder().decode(VerifyResponse.self, from: data)
}

private func handleVerifyResult(_ result: VerifyResponse) async {
    if result.valid {
        await setState(.activated)
        await setPlan(LicensePlan(rawValue: result.plan) ?? .unknown)
        UserDefaults.standard.set(Date(), forKey: "com.xlsone.license.lastValid")
    } else if result.expired == true {
        await setState(.expired)
    } else {
        await setState(.unactivated)
    }
}

private func postJSON(to urlString: String, body: [String: String]) async -> Data? {
    guard let url = URL(string: urlString) else { return nil }

    var request = URLRequest(url: url, timeoutInterval: API.timeout)
    request.httpMethod = "POST"
    request.setValue("application/json", forHTTPHeaderField: "Content-Type")
    request.httpBody = try? JSONSerialization.data(withJSONObject: body)
}

```

```

do {
    let (data, response) = try await URLSession.shared.data(for: request)
    guard let http = response as? HTTPURLResponse, (200...299).contains(http.statusCode) else {
        return nil
    }
    return data
} catch {
    return nil
}
}

// MARK: - Private — Offline

private func tryOfflineActivation(key: String, deviceID: String) -> ActivationResult? {
    // Offline activation requires a pre-generated .license file
    guard let licenseData = loadOfflineLicense() else { return nil }

    // Verify the offline license matches the key and device
    guard let payload = decodeTokenPayload(licenseData),
        payload.sub == key,
        payload.dev == deviceID,
        isTokenValidLocally(payload, deviceID: deviceID)
    else { return nil }

    saveToken(licenseData)
    let plan = LicensePlan(rawValue: payload.plan) ?? .personalLifetime
    Task { @MainActor in
        self.licenseState = .activated
        self.plan = plan
    }
    return .success(plan: plan, expiresAt: payload.exp > 0
        ? Date(timeIntervalSince1970: TimeInterval(payload.exp)).ISO8601Format() : nil)
}

private func isTokenValidLocally(_ payload: TokenPayload, deviceID: String) -> Bool {
    guard payload.dev == deviceID else { return false }
    // Lifetime token
    if payload.exp == 0 { return true }
    // Subscription token with expiry
    return payload.exp > Int(Date().timeIntervalSince1970)
}

// MARK: - Private — Storage

private func loadPersistedState() {
    if let token = loadToken(), let payload = decodeTokenPayload(token),
        isTokenValidLocally(payload, deviceID: getOrCreateDeviceID()) {
        licenseState = .activated
        plan = LicensePlan(rawValue: payload.plan) ?? .unknown
        return
    }
    let remaining = checkTrialStatus()
    if remaining > 0 {
        licenseState = .trial(remainingDays: remaining)
    }
}

private func getOrCreateDeviceID() -> String {
    if let existing = Keychain.load(key: Storage.deviceIDKey) {
        return existing
    }
}

```

```

}

let id = DeviceFingerprint.generate()
_ = Keychain.save(key: Storage.deviceIDKey, data: id)
return id
}

private func loadToken() -> String? {
    Keychain.load(key: Storage.tokenKey)
}

private func saveToken(_ token: String) {
    _ = Keychain.save(key: Storage.tokenKey, data: token)
}

private func loadOfflineLicense() -> String? {
    UserDefaults.standard.string(forKey: Storage.offlineLicenseKey)
}

private func decodeTokenPayload(_ token: String) -> TokenPayload? {
    let parts = token.split(separator: ".")
    guard parts.count >= 2 else { return nil }
    // Base64URL decode the payload (second segment)
    var base64 = String(parts[1])
        .replacingOccurrences(of: "-", with: "+")
        .replacingOccurrences(of: "_", with: "/")
    while base64.count % 4 != 0 { base64 += "=" }
    guard let data = Data(base64Encoded: base64),
        let payload = try? JSONDecoder().decode(TokenPayload.self, from: data)
    else { return nil }
    return payload
}

@MainActor private func setState(_ state: LicenseState) { self.licenseState = state }
@MainActor private func setPlan(_ plan: LicensePlan) { self.plan = plan }

#if os(macOS)
@MainActor
private static func runCenteredModal(_ panel: NSSavePanel) -> NSApplication.ModalResponse {
    panel.contentView?.layoutSubtreeIfNeeded()
    if let owner = NSApp.keyWindow ?? NSApp.mainWindow ?? NSApp.windows.first(where: { $0.isVisible }) {
        let ownerFrame = owner.frame
        let panelFrame = panel.frame
        panel.setFrameOrigin(NSPoint(
            x: ownerFrame.midX - panelFrame.width / 2,
            y: ownerFrame.midY - panelFrame.height / 2
        ))
    } else {
        panel.center()
    }
    return panel.runModal()
}
#endif

private func mapAPIError(_ error: APIError) -> ActivationError {
    switch error.error {
    case "KEY_NOT_FOUND":    return .keyNotFound
    case "KEY_REVOKED":     return .keyRevoked
    case "DEVICE_LIMIT":    return .deviceLimit
    case "RATE_LIMITED":    return .rateLimited
    default:                return .serverError(error.message ?? LocaleManager.loc("未知错误"))
    }
}

```

```

}
}
}

// MARK: - Types

public enum LicenseState: Equatable {
    case unactivated
    case activated
    case expired
    case gracePeriod // Offline, token not verified recently but within grace window
    case trial(remainingDays: Int) // Free trial with N days remaining
}

public enum LicensePlan: String {
    case appStore = "app_store"
    case personalYearly = "personal_yearly"
    case personalLifetime = "personal_lifetime"
    case enterprise10 = "enterprise_10"
    case enterprise25 = "enterprise_25"
    case enterpriseUnlimited = "enterprise_unlimited"
    case unknown = "unknown"

    public var displayName: String {
        switch self {
            case .appStore:          return LocaleManager.loc("App Store")
            case .personalYearly:    return LocaleManager.loc("个人年度订阅")
            case .personalLifetime:  return LocaleManager.loc("个人永久授权")
            case .enterprise10:      return LocaleManager.loc("企业版 (10 台)")
            case .enterprise25:      return LocaleManager.loc("企业版 (25 台)")
            case .enterpriseUnlimited: return LocaleManager.loc("企业版 (无限)")
            case .unknown:          return LocaleManager.loc("未知")
        }
    }
}

public enum ActivationResult: Equatable {
    case success(plan: LicensePlan, expiresAt: String?)
    case failure(ActivationError)
}

public enum ActivationError: Error, Equatable {
    case invalidKeyFormat
    case invalidLicenseFile
    case licenseDeviceMismatch
    case keyNotFound
    case keyRevoked
    case deviceLimit
    case networkError
    case rateLimited
    case serverError(String)

    public var localizedDescription: String {
        switch self {
            case .invalidKeyFormat: return LocaleManager.loc("激活码格式不正确")
            case .invalidLicenseFile: return LocaleManager.loc("授权文件无效或已过期")
            case .licenseDeviceMismatch: return LocaleManager.loc("授权文件与当前设备不匹配")
            case .keyNotFound: return LocaleManager.loc("激活码不存在")
            case .keyRevoked: return LocaleManager.loc("激活码已被吊销")
            case .deviceLimit: return LocaleManager.loc("已达到最大设备数限制")
        }
    }
}

```

```

case .networkError:    return LocaleManager.loc("无法连接激活服务器, 请检查网络")
    case .rateLimited:  return LocaleManager.loc("请求过于频繁, 请稍后再试")
    case .serverError(let m): return LocaleManager.loc("服务器错误: \(m)")
    }
}

// MARK: - API Response Types (internal)

struct ActivateResponse: Decodable {
    let license_token: String
    let plan: String
    let expires_at: String?
    let device_id: String
}

struct VerifyResponse: Decodable {
    let valid: Bool
    let plan: String
    let expires_at: String?
    let expired: Bool?

    enum CodingKeys: String, CodingKey {
        case valid, plan, expires_at
    }

    init(from decoder: Decoder) throws {
        let container = try decoder.container(keyedBy: CodingKeys.self)
        valid = try container.decode(Bool.self, forKey: .valid)
        plan = (try? container.decode(String.self, forKey: .plan)) ?? "unknown"
        expires_at = try? container.decodeIfPresent(String.self, forKey: .expires_at)
        expired = nil // derived from error code if valid=false
    }
}

struct RefreshResponse: Decodable {
    let valid: Bool
    let token: String?
    let refreshed: Bool?
}

struct APIError: Decodable {
    let error: String
    let message: String?
}

struct TokenPayload: Decodable {
    let sub: String // key_id
    let dev: String // device_id
    let plan: String
    let iat: Int
    let exp: Int // 0 = lifetime
}

// MARK: - Key Format Validation

enum KeyFormat {
    static let pattern = try! NSRegularExpression(pattern: "[A-Z0-9]{4}-[A-Z0-9]{4}-[A-Z0-9]{4}$")

    static func isValid(_ key: String) -> Bool {

```

```

pattern.firstMatch(in: key, range: NSRange(key.startIndex..., in: key)) != nil
    }
}

// File: Sources/xlsOneLicense/LicenseActivationView.swift
import SwiftUI
import xlsOneCore

// MARK: - License Activation View

/// Modal sheet shown when the app needs activation.
public struct LicenseActivationView: View {
    @ObservedObject var licenseManager: LicenseManager
    @ObservedObject private var localeManager = LocaleManager.shared
    @State private var keyInput = ""
    @State private var isActivating = false
    @State private var errorMessage: String?
    @State private var successMessage: String?
    @State private var showOfflineInfo = false

    public init(licenseManager: LicenseManager = .shared) {
        self.licenseManager = licenseManager
    }

    public var body: some View {
        VStack(spacing: 0) {
            // Icon + Title
            VStack(spacing: 12) {
                Image(systemName: "key.fill")
                    .font(.system(size: 40))
                    .foregroundColor(.accentColor)

                Text(LocaleManager.loc("激活 表表归一"))
                    .font(.title2)
                    .fontWeight(.semibold)

                if licenseManager.licenseState == .expired {
                    Text(LocaleManager.loc("您的许可证已过期，请续费获取新的激活码"))
                        .font(.subheadline)
                        .foregroundColor(.secondary)
                        .multilineTextAlignment(.center)
                } else {
                    Text(LocaleManager.loc("输入激活码以解锁全部功能"))
                        .font(.subheadline)
                        .foregroundColor(.secondary)
                }
            }
            .padding(.top, 32)
            .padding(.bottom, 24)

            // Activation Key Input
            VStack(alignment: .leading, spacing: 8) {
                Text(LocaleManager.loc("激活码"))
                    .font(.callout)
                    .fontWeight(.medium)

                TextField("XXXX-XXXX-XXXX", text: $keyInput)
                    .textFieldStyle(.plain)
                    .font(.system(.body, design: .monospaced))
                    .padding(12)
            }
        }
    }
}

```



```

.background(Color(nsColor: .controlBackgroundColor))
    .clipShape(RoundedRectangle(cornerRadius: 8))
    .overlay(
        RoundedRectangle(cornerRadius: 8)
            .stroke(Color.secondary.opacity(0.3), lineWidth: 1)
    )
    .onChange(of: keyInput) { newValue in
        // Auto-insert dashes and uppercase
        var cleaned = newValue.uppercased().replacingOccurrences(of: "-", with: "")
        if cleaned.count > 12 { cleaned = String(cleaned.prefix(12)) }

        var formatted = ""
        for (i, ch) in cleaned.enumerated() {
            if i > 0 && i % 4 == 0 { formatted += "-" }
            formatted.append(ch)
        }
        if formatted != newValue {
            keyInput = formatted
        }
    }
}

// Error message
if let error = errorMessage {
    HStack(spacing: 6) {
        Image(systemName: "exclamationmark.triangle.fill")
            .font(.caption)
        Text(error)
            .font(.caption)
    }
    .foregroundColor(.red)
    .padding(.top, 8)
}

if let success = successMessage {
    HStack(spacing: 6) {
        Image(systemName: "checkmark.circle.fill")
            .font(.caption)
        Text(success)
            .font(.caption)
    }
    .foregroundColor(.green)
    .padding(.top, 8)
}

// Spacer
Spacer().frame(height: 20)

// Activate Button
Button(action: activate) {
    HStack(spacing: 8) {
        if isActivating {
            ProgressView()
                .scaleEffect(0.8)
        }
        Text(isActivating ? LocaleManager.loc("验证中...") : LocaleManager.loc("激活"))
            .fontWeight(.semibold)
    }
}
.frame(maxWidth: .infinity)
.padding(.vertical, 12)

```

```

.background(validInput ? Color.accentColor : Color.gray.opacity(0.3))
    .foregroundColor(validInput ? .white : .secondary)
    .clipShape(RoundedRectangle(cornerRadius: 8))
}
.disabled(!validInput || isActivating)
.buttonStyle(.plain)

Spacer().frame(height: 12)

// Trial & Purchase
HStack(spacing: 24) {
    Button(LocaleManager.loc("免费试用 14 天")) {
        let remaining = licenseManager.startTrial()
        if remaining > 0 {
            errorMessage = nil
            successMessage = nil
            licenseManager.showActivationSheet = false
        }
    }
    .buttonStyle(.link)

    Button(LocaleManager.loc("购买激活码 →")) {
        if let url = URL(string: "https://z-pulse.cn") {
            NSWorkspace.shared.open(url)
        }
    }
    .buttonStyle(.link)
}
.font(.subheadline)

Spacer().frame(height: 12)

// Offline activation
Button {
    showOfflineInfo.toggle()
} label: {
    HStack(spacing: 4) {
        Image(systemName: "wifi.slash")
        Text(LocaleManager.loc("离线激活"))
    }
    .font(.caption)
    .foregroundColor(.secondary)
}
.buttonStyle(.plain)

if showOfflineInfo {
    VStack(alignment: .leading, spacing: 4) {
        Text(LocaleManager.loc("离线激活方式: "))
            .font(.caption)
            .fontWeight(.medium)
        Text(LocaleManager.loc("1. 在互联网电脑上访问 z-pulse.cn/offline"))
            .font(.caption)
            .foregroundColor(.secondary)
        Text(LocaleManager.loc("2. 输入购买邮箱和本机设备码"))
            .font(.caption)
            .foregroundColor(.secondary)
        Text(LocaleManager.loc("3. 下载授权文件并导入本程序"))
            .font(.caption)
            .foregroundColor(.secondary)
    }
}

```

```

Button {
    importLicenseFile()
    } label: {
        Label(LocaleManager.loc("导入授权文件..."), systemImage: "doc.badge.plus")
        .frame(maxWidth: .infinity)
    }
    .buttonStyle(.bordered)
    .padding(.top, 8)
}
.padding(12)
.background(Color(nsColor: .controlBackgroundColor))
.clipShape(RoundedRectangle(cornerRadius: 8))
}

Spacer()
}
.padding(.horizontal, 48)
.frame(width: 420, height: 520)
.background(Color(nsColor: .windowBackgroundColor))
}

private var validInput: Bool {
    keyInput.count == 14 // XXXX-XXXX-XXXX
}

private func activate() {
    guard validInput else { return }
    isActivating = true
    errorMessage = nil
    successMessage = nil

    Task {
        let result = await licenseManager.activate(key: keyInput)
        await MainActor.run {
            isActivating = false
            switch result {
            case .success:
                licenseManager.showActivationSheet = false
            case .failure(let error):
                errorMessage = error.localizedDescription
            }
        }
    }
}

private func importLicenseFile() {
    errorMessage = nil
    successMessage = nil

    guard let result = licenseManager.importOfflineLicenseFile() else { return }
    switch result {
    case .success(let plan, _):
        successMessage = LocaleManager.loc("授权文件已导入: %@").replacingOccurrences(of: "%@", with: plan.displayName)
        licenseManager.showActivationSheet = false
    case .failure(let error):
        errorMessage = error.localizedDescription
    }
}
}

```

```

// MARK: - License Status Badge

/// Small indicator showing license status in toolbar
public struct LicenseStatusBadge: View {
    @ObservedObject var licenseManager: LicenseManager

    public init(licenseManager: LicenseManager = .shared) {
        self.licenseManager = licenseManager
    }

    public var body: some View {
        HStack(spacing: 4) {
            Circle()
                .fill(color)
                .frame(width: 6, height: 6)
            Text(label)
                .font(.caption)
                .foregroundColor(.secondary)
        }
    }

    private var color: Color {
        switch licenseManager.licenseState {
        case .activated: return .green
        case .expired,
             .unactivated: return .red
        case .gracePeriod: return .orange
        case .trial: return .blue
        }
    }

    private var label: String {
        switch licenseManager.licenseState {
        case .activated: return licenseManager.plan.displayName
        case .unactivated: return LocaleManager.loc("未激活")
        case .expired: return LocaleManager.loc("已过期")
        case .gracePeriod: return LocaleManager.loc("离线模式")
        case .trial(let remaining): return String(format: LocaleManager.loc("试用 %lld 天"), remaining)
        }
    }
}

```